

Chapter 10

Moments of Inertia

Area moments of inertia are a measure of the distribution of a two-dimensional area around a particular axis. Fundamentally, the portions of a shape which are located farther from the axis have a greater affect than the parts which are closer. The primary application is in structural engineering and machine design where they are used to determine a structural member's stiffness. Another application is in Fluid Mechanics where they are used to determine the effect of pressure on a submerged surface. We will use the symbol I for this property, along with a subscript to indicate the specific axis, so for example, I_x would indicate the “Area moment of inertia with respect to the x axis.”

Warning 10.0.1

The *mass* moment of inertia you learned about in Physics is not the same as the *area* moment of inertia in Statics!

This can be confusing since both are commonly shortened to “moment of inertia” and both use the same symbol, I . They have different units however, and the intended moment of inertia can easily be determined from context or with a unit analysis.

10.1 Integral Properties of Shapes

Key Questions

- Why does the area moment of inertia integral equation include a distance squared term?
- When performing a single integral, either dx or dy , what is your differential element dA shape?

As you know, two dimensional shapes like rectangles and circles have properties such as area, perimeter, and centroid. These are purely *geometric* properties

since they belong to the shape alone, in contrast to *physical* properties like weight and mass which belong to real physical objects.

In this section we introduce several new geometric properties useful in engineering including the **Area Moment of Inertia**.

The integral properties of shapes, along with the names and symbols commonly used to represent them are given in the table below. You are already familiar with *area* from Geometry and the *first moment of area* from Chapter 7. The remaining properties are the subject of this chapter. They all have a similar form, and can be evaluated using similar integration techniques.

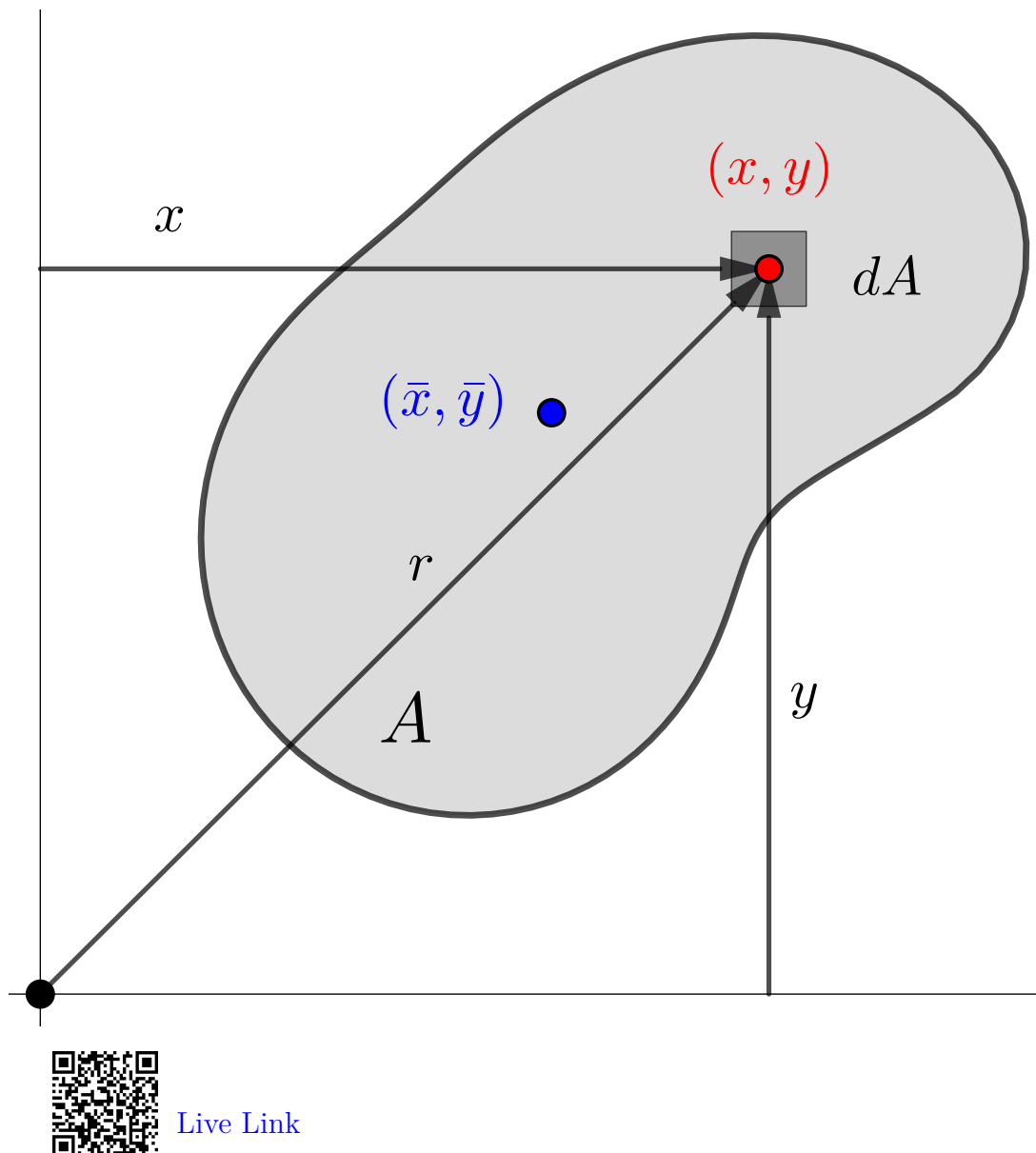


Figure 10.1.1 Definitions for area properties

This interactive diagram defines the terms which apply to all of the property

definitions below. It shows a generic plane area A , divided into differential elements $dA = dx \, dy$. The differential element dA is an infinitesimally tiny rectangle centered about point (x, y) , which can range over the entire area. The distances to the element from the y axis, the x axis and the origin are designated x , y , and r respectively. The centroid of the entire area is located at $(\bar{x}, \bar{y})$.

Definition	Name	More Information
$A = \int_A dA$	Area	Subsection 7.7.2
$Q_x = \int_A y \, dA$	First Moment of Area (with respect to the x axis)	Chapter 7
$Q_y = \int_A x \, dA$	First Moment of Area (with respect to the y axis)	Chapter 7
$I_x = \int_A y^2 \, dA$	Second Moment of Area, or Moment of Inertia (with respect to the x axis)	Section 10.2
$I_y = \int_A x^2 \, dA$	Second Moment of Area, or Moment of Inertia (with respect to the y axis)	Section 10.2
$J_O = \int_A r^2 \, dA$	Polar Moment of Inertia	Section 10.5
$I_{xy} = \int_A x \, y \, dA$	Product of Inertia	Section 10.7

All of these properties are defined as integrals over an area A . These integrals may be evaluated by double-integrating over x and y in Cartesian coordinates or r and θ in polar coordinates. They can also be evaluated using single integration using the methods demonstrated in [Subsection 10.2.2](#).

None of these integrals can be evaluated until a specific shape is chosen. When shape has been specified, the bounding functions and integration limits can be determined and only then may the integral be solved using appropriate integration techniques. If the shape is specified in general terms, say a rectangle with base b and height h , then the result of the integration will be a formula for the property applicable to all similar shapes.

10.1.1 Area

The total **area** of a shape is found by integrating the differential elements of area over the entire shape.

$$A = \int_A dA. \quad (10.1.1)$$

The limit on this integral is indicated with an A to indicate that the integration is carried out over the entire area. The resulting value will have units of $[\text{length}]^2$ and does not depend on the position of the shape on the coordinate plane.

Since the area formulas for common shapes are well known, you only need to use integration in uncommon situations.

10.1.2 First Moment of Area

The **first moment of area**, which was introduced in [Chapter 7](#), is defined by these two equations.

$$Q_x = \int_A y \, dA \qquad Q_y = \int_A x \, dA \quad (10.1.2)$$

and has units of $[\text{length}]^3$.

The first moment of area with respect to an axis is a measure of the distribution of the shape about an axis. It depends on the shape and also its location on the coordinate plain. Portions of area on the negative side of the selected axis make the first moment smaller, while areas on the positive side make it larger. If the shape's centroid is located exactly on the axis, the integral will sum to zero because the contributions of area above and below the axis cancel each other.

The average value of the first moment of area is found by is found by dividing the first moment by the area of the shape, and the result indicates the distance from the axis to centroid of the shape.

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{\sum A_i} \quad \bar{y} = \frac{\sum \bar{y}_i A_i}{\sum A_i}.$$

10.1.3 Moment of Inertia

The **area moment of inertia**, the subject of this chapter, is defined by these two equations.

$$I_x = \int_A y^2 dA \qquad I_y = \int_A x^2 dA \qquad (10.1.3)$$

and has units of $[\text{length}]^4$.

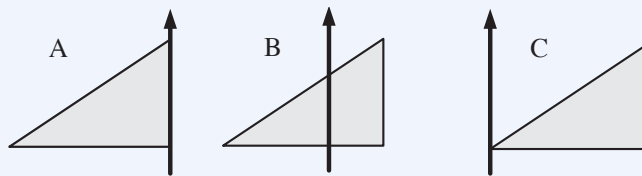
As you can see, these equations are similar to the equations for the [first moment of area \(10.1.2\)](#), except that the distance terms x and y are now squared. In recognition of the similarity, the area moments of inertia are also known as the **second moments of area**. We will use the terms moment of inertia and second moment interchangeably. These two quantities are sometimes designated as *rectangular* moments of inertia to distinguish them from the polar moment of inertia described in the next section.

Like the first moment, the second moment of area provides a measure of the distribution of area around an axis, but in this case the distance to each element is squared. This gives increased importance to portions of the area which are far from the axis. Squaring the distance means that identical elements on opposite sides of the axis *both* contribute to the sum rather than cancel each other out as they do in the first moment. As a result, the moment of inertia is always a positive quantity.

Two identical shapes can have completely different moments of inertia, depending on how the shape is distributed around the axis. A shape with most of its area close to the axis has a smaller moment of inertia than the same shape would if its area was distributed farther from the axis. This is a non-linear effect, because when the distance term is doubled, the contribution of that element to the sum increases fourfold.

Question 10.1.2

These three triangles are all the same size. Rank them from smallest to largest moment of inertia with respect to the y axis.



Answer. From smallest to largest: $I_B < I_A < I_C$.

Solution. Although the areas of all three triangles are the same, triangle B has area on both sides of the y axis and relatively close to it, and so has the smallest moment of inertia, while triangle C has the most of its area far from the y axis which makes its moment of inertia largest.

For parallel axes, the minimum moment of inertia will be about an axis

passing through the centroid, and when axis B is centroidal $I_C = 3I_A = 9I_B$.

Moving a shape away from the axis (or moving the axis away from the shape) increases its moment of inertia, and moving it closer to the axis decreases it, until it crosses to the other side of the axis, and then its moment of inertia will begin to increase again.

The minimum moment of inertia occurs when the centroid of the shape falls on the axis. When this occurs, the moment of inertia is called the **centroidal moment of inertia**. A bar over the symbol I is used to indicate that a moment of inertia is centroidal. So for example, $\bar{I}_x$ and $\bar{I}_y$ represent the “centroidal moment of inertia with respect to the x axis” and the “the centroidal moment of inertia with respect to the y axis.” The bar in this case does not mean that moment of inertia is a vector quantity. Note that a shape can have multiple centroidal moments of inertia, because more than one axis can pass through the centroid of a shape. In this text, we will only the vertical and horizontal axes, but they are not the only possibilities.

The centroidal moment of inertia is particularly important. We will see in (10.3.1) that if we know a shape’s centroidal moment of inertia for some axis direction, it is a simple process to calculate the moment of inertia of the shape about any other parallel axis. The moment of inertia is used in Mechanics of Materials to find stress and deflection in beams and to determine the load which will cause a column to buckle.

We stated earlier that the centroidal moment of inertia is the minimum moment of inertia, but by this we mean, the minimum moment of inertia for a particular axis direction, for example horizontal. Other centroidal axes may have a different moment of inertia, either larger or smaller than the moment of inertia about a horizontal centroidal axis. The centroidal axes which have the absolute minimum and maximum moment of inertia are called the principle axes. The principle axes are not necessarily horizontal and vertical.

Thinking Deeper 10.1.3 Beam bending.

To get a feel for how moment of inertia affects engineering design, find a ruler, a yardstick, or something similar: long with a rectangular cross section.

Try to bend the ruler both when it’s flat and also when it’s turned on edge. You will find that bending the ruler around the x axis while it’s flat is easy compared to bending it the other way, around the z axis. Why is it easier to bend the ruler one way than the other? It’s the same object, made of the same material either way.

The answer has to do with the moment of inertia, and how it relates to the bending axis.

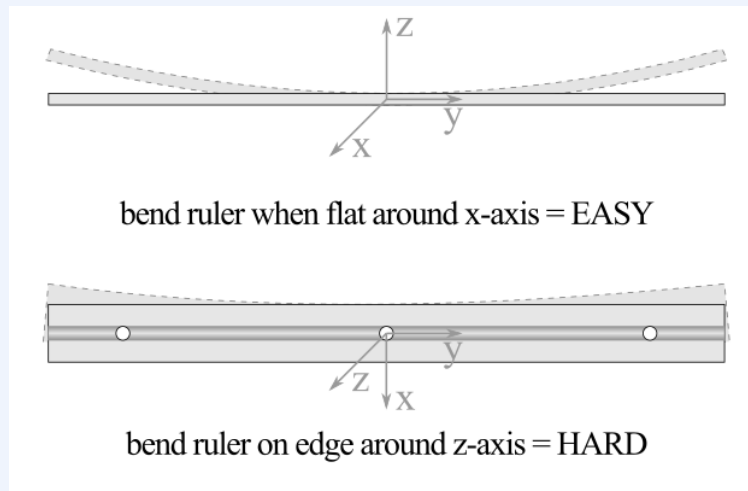


Figure 10.1.4 Bending a ruler.

As engineers we are not satisfied with merely knowing that it's harder to bend a ruler one way than the other, we'd like to know *how much harder*? For a 1/8 in thick ruler that is 1 in tall, the bending resistance about the z axis is over 20 times more than the bending resistance the other way, about the x axis.

To further see how the moment of inertia comes into play, consider the curvature caused by applying opposing moments to the ends of a beam such as your ruler. You will cause it to bend into an arc of a circle of some radius. A curious engineer would like to know how the curvature of the beam is related to the applied moment, the geometry, and the physical properties of the beam.

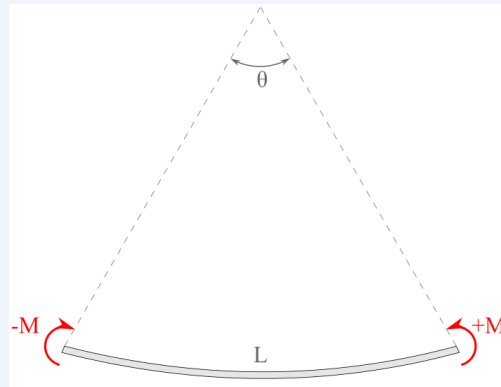


Figure 10.1.5 Beam of length L which is being bent by opposing couple-moments to an arc with angle θ .

You will learn in Mechanics of Materials that the relationship is:

$$M = \theta \left(\frac{EI}{L} \right) \quad (10.1.4)$$

where:

E is a material property called Young's Modulus or the modulus of elasticity which characterizes the stiffness of a material.

L is the length of the beam, and

I is the moment of inertia of the cross-section of the beam about the bending axis.

M is the moment applied to the ends of the beam, and

θ is the curvature of the beam.

Since E and I are in the numerator and L is in the denominator, a longer beam is more flexible and larger values of E or I make the beam stiffer. With those properties fixed, angle θ is directly proportional to the moment M .

The sag, or **deflection**, of a beam when supporting a load is also related to these factors, and the placement of the load as well. For example, if a beam is loaded with a concentrated force P at its center its maximum deflection $\delta_{\max}$ will occur at the midpoint, with

$$\delta_{\max} = \frac{PL^3}{48EI}$$

10.1.4 Polar Moment of Inertia

The **polar moment of inertia** is defined as

$$J_O = \int_A r^2 dA \quad (10.1.5)$$

and has units of $[\text{length}]^4$.

The polar moment of inertia is another measure of the distribution of an area but, in this case, about a point at the origin rather than about an axis. One important application of this value is to quantify the resistance of a shaft to torsion or twisting due to the shape of its cross-section.

Thinking Deeper 10.1.6 Why don't we call the polar moment of inertia I_z ?

The squared distance in the polar moment of inertia formula is the distance from the z axis, so it would seem reasonable to name the polar moment I_z to be consistent with I_x and I_y , which use distances from the x and y axes.

Instead engineers use the letter J to represent this quantity. Why?

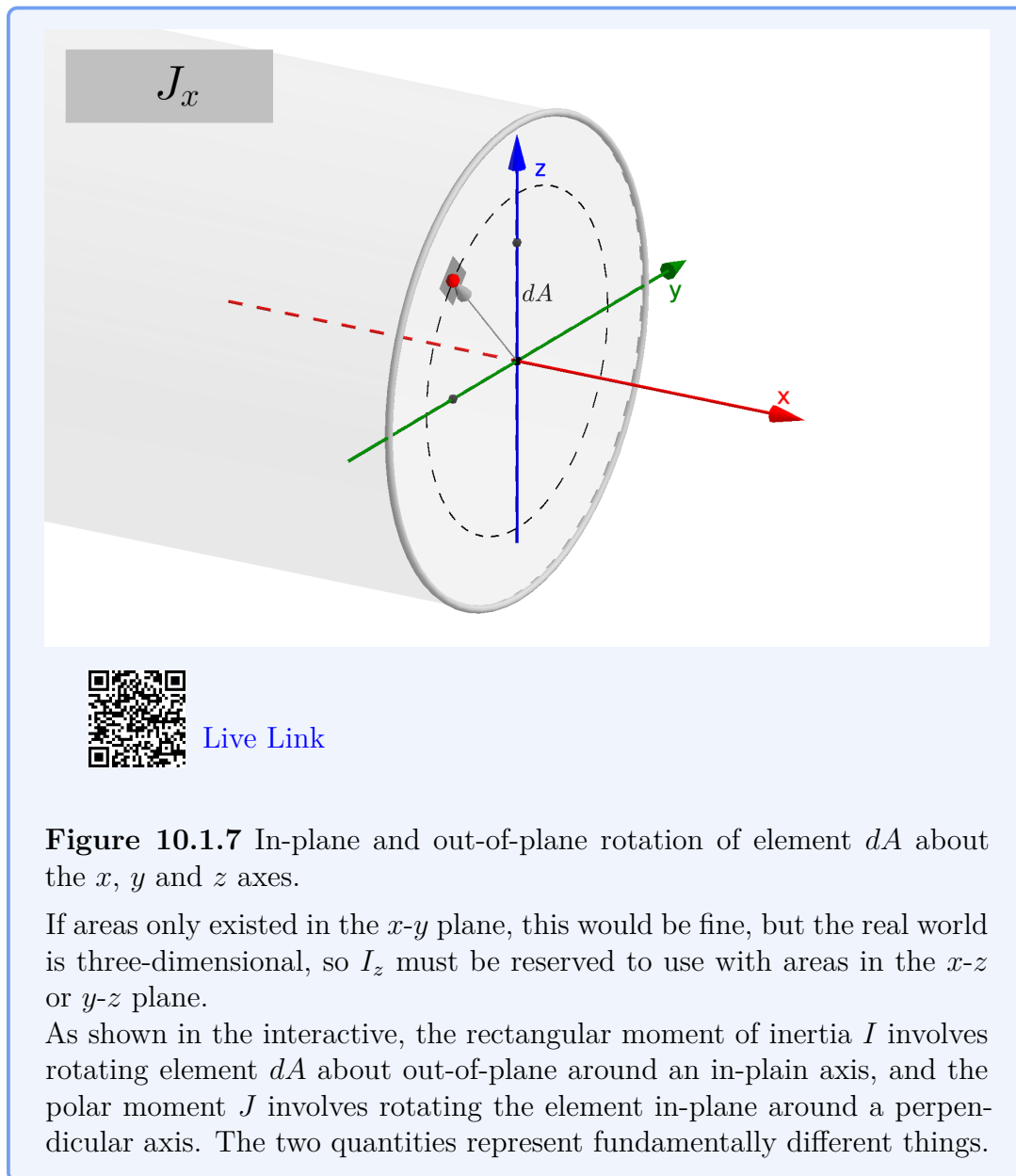


Figure 10.1.7 In-plane and out-of-plane rotation of element dA about the x , y and z axes.

If areas only existed in the x - y plane, this would be fine, but the real world is three-dimensional, so I_z must be reserved to use with areas in the x - z or y - z plane.

As shown in the interactive, the rectangular moment of inertia I involves rotating element dA about out-of-plane around an in-plane axis, and the polar moment J involves rotating the element in-plane around a perpendicular axis. The two quantities represent fundamentally different things.

10.1.5 Product of Inertia

The final property of interest is the **product of inertia** and it is defined as

$$I_{xy} = \int_A xy \, dA \quad (10.1.6)$$

where x and y are defined as in Figure 10.1.1. Like the others, the units associated with this quantity are $[\text{length}]^4$. The name was chosen because the distance squared term in the integral is the *product* of the element's coordinates. In contrast to the other area moments, which are always positive, the product of inertia can be a positive, negative or zero.

10.2 Moments of Inertia of Common Shapes

In following sections we will use the integral definitions of moment of inertia (10.1.3) to find the moments of inertia of five common shapes: rectangle, triangle, circle, semi-circle, and quarter-circle with respect to a specified axis. The integration techniques demonstrated can be used to find the moment of inertia of any two-dimensional shape about any desired axis.

Moments of inertia depend on both the shape, and the axis. Pay attention to the placement of the axis with respect to the shape, because if the axis is located elsewhere or oriented differently, the results will be different.

We will begin with the simplest case: the moment of inertia of a rectangle about a horizontal axis located at its base. This case arises frequently and is especially simple because the boundaries of the shape are all constants.

10.2.1 Moment of Inertia of a Rectangle

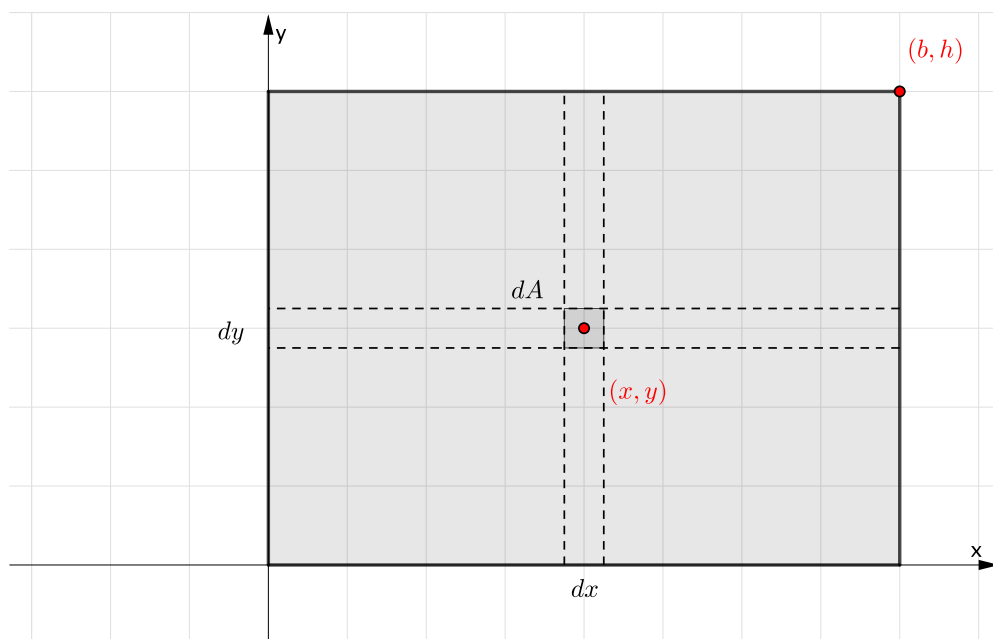
Consider the $(b \times h)$ rectangle shown. This rectangle is oriented with its bottom-left corner at the origin and its upper-right corner at the point (b, h) , where b and h are constants.

What is the moment of inertia of this rectangle with respect to the x axis?

To find the moment of inertia, divide the area into square differential elements dA at (x, y) where x and y can range over the entire rectangle and then evaluate the integral using double integration.

The differential element dA has width dx and height dy , so

$$dA = dx \, dy = dy \, dx. \quad (10.2.1)$$



[Live Link](#)

Figure 10.2.1 Rectangle divided into vertical and horizontal differential strips

It would seem like this is an insignificant difference, but the order of dx and dy in this expression determines the order of integration of the double integral. We will try both ways and see that the result is identical.

Using $dA = dx \, dy$

First, we will evaluate (10.1.3) using $dA = dx \, dy$.

If you are not familiar with double integration, briefly you can think of a double integral as two normal single integrals, one ‘inside’ and the other ‘outside,’ which are evaluated one at a time from the inside out. Our integral becomes

$$\begin{aligned}
 I_x &= \int_A y^2 dA \\
 &= \iint y^2 \underbrace{dx \, dy}_{dA} \\
 &= \underbrace{\int_{\text{bottom}}^{\text{top}} \left[\underbrace{\int_{\text{left}}^{\text{right}} y^2 dx}_{\text{inside}} \right] dy}_{\text{outside}}
 \end{aligned}$$

The limits on double integrals are usually *functions* of x or y , but for this rectangle the limits are all *constants*. The bottom and top limits are $y = 0$ and $y = h$; the left and right limits are $x = 0$ and $x = b$. Note that the y^2 term can be taken out of the inside integral, because in terms of x , it is constant.

Inserting $dx \, dy$ for dA and the limits into (10.1.3), and integrating starting with the inside integral gives

$$\begin{aligned}
 I_x &= \int_A y^2 dA \\
 &= \int_0^h \int_0^b y^2 \, dx \, dy \\
 &= \int_0^h y^2 \int_0^b dx \, dy \\
 &= \int_0^h y^2 \boxed{b \, dy} \\
 &= b \int_0^h y^2 \, dy \\
 &= b \left. \frac{y^3}{3} \right|_0^h
 \end{aligned}$$

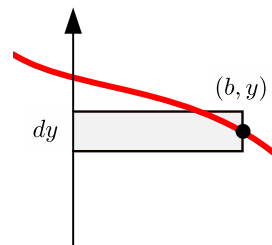
Evaluating the limit gives the result

$$I_x = \frac{bh^3}{3}. \quad (10.2.2)$$

This is the formula for the moment of inertia of a rectangle about an axis passing through its base, and is worth remembering.

The boxed quantity is the *result* of the inside integral times dx , and can be interpreted as the *differential area* of a horizontal strip,

$$dA = b \, dy.$$



This will allow us to set up a problem as a single integral using strips and skip the inside integral completely as we will see in [Subsection 10.2.2](#).

This result means that the moment of inertia of the rectangle depends only on the dimensions of the base and height and has units $[\text{length}]^4$. The height term is cubed and the base is not, which is unsurprising because the moment of inertia gives more importance to parts of the shape which are farther away from the axis. Doubling the width of the rectangle will double I_x but doubling the height will increase I_x eightfold. In all moment of inertia formulas, the dimension perpendicular to the axis is always cubed.

Warning 10.2.2

This result is for this particular situation; you will get a different result for a different shape or a different axis.

Using $dA = dy \, dx$

Now, we will evaluate (10.1.3) using $dA = dy \, dx$ which reverses the order of integration and means that the integral over y gets conducted first. Since the distance-squared term y^2 is a function of y it remains inside the inside integral this time and the result of the inside integral is not an area as it was previously.

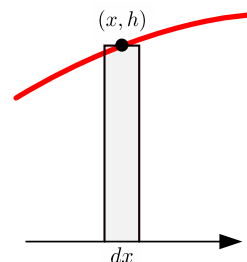
Inserting $dy \, dx$ for dA and the limits into (10.1.3), and integrating gives

$$\begin{aligned}
 I_x &= \int_A y^2 \, dA \\
 &= \int_0^b \int_0^h y^2 \, dy \, dx \\
 &= \int_0^b \left. \frac{y^3}{3} \right|_0^h dx \\
 &= \int_0^b \boxed{\frac{h^3}{3} \, dx} \\
 &= \frac{h^3}{3} \int_0^b dx \\
 I_x &= \frac{bh^3}{3}
 \end{aligned}$$

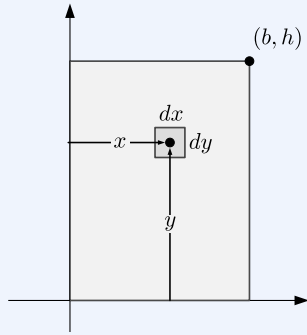
As before, the result is the moment of inertia of a rectangle with base b and height h , about an axis passing through its base. We have found that the moment of inertia of a rectangle about an axis through its base is (10.2.2), the same as before.

The boxed quantity is the *result* of the inside integral times dx , and can be interpreted as the *differential moment of inertia* of a vertical strip about the x axis. This is consistent our previous result. The vertical strip has a base of dx and a height of h , so its moment of inertia by (10.2.2) is

$$dI_x = \frac{h^3}{3} dx. \quad (10.2.3)$$



We will use these results to set up problems as a single integral which sum the moments of inertia of the differential strips which cover the area in Subsection 10.2.3.

Example 10.2.3 I_y of a Rectangle.

Find the moment of inertia of the rectangle about the y axis using square differential elements dA .

Solution 1. Following the same procedure as before, we divide the rectangle into square differential elements $dA = dx \, dy$ and evaluate the double integral for I_y from (10.1.3) first by integrating over x , and then over y .

$$\begin{aligned}
 I_y &= \int_A x^2 dA \\
 &= \int_0^h \int_0^b x^2 \, dx \, dy \\
 &= \int_0^h \left[\int_0^b x^2 \, dx \right] dy \\
 &= \int_0^h \left[\frac{x^3}{3} \right]_0^b dy \\
 &= \int_0^h \left[\frac{b^3}{3} \right] dy \\
 &= \frac{b^3}{3} y \Big|_0^h \\
 I_y &= \frac{b^3 h}{3}
 \end{aligned}$$

The formula for I_y is the same as the formula as we found previously for I_x except that the base and height terms have reversed roles. Here, the horizontal dimension is cubed and the vertical dimension is the linear term. In all moment of inertia formulas, the dimension perpendicular to the axis is cubed.

Solution 2. This solution demonstrates that the result is the same when the order of integration is reversed. This time we evaluate I_y by dividing the rectangle into square differential elements $dA = dy \, dx$ so the inside integral is now with respect to y and the outside integral is with respect

to x .

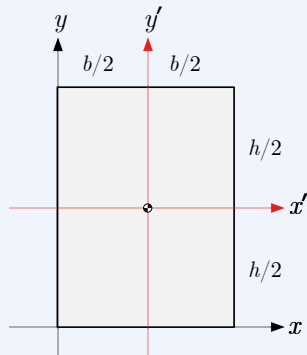
$$\begin{aligned}
 I_y &= \int_A x^2 dA \\
 &= \int_0^b x^2 \left[\int_0^h dy \right] dx \\
 &= \int_0^b x^2 [h dx] \\
 &= h \int_0^b x^2 dx \\
 &= h \left. \frac{x^3}{3} \right|_0^b \\
 I_y &= \frac{hb^3}{3}
 \end{aligned}$$

Centroidal Moment of Inertia

As discussed in [Subsection 10.1.3](#), a moment of inertia about an axis passing through the area's centroid is a *Centroidal* Moment of Inertia. The convention is to place a bar over the symbol I when the axis is centroidal.

The following example finds the centroidal moment of inertia for a rectangle using integration.

Example 10.2.4 Rectangle.



Use integration to find the moment of inertia of a $(b \times h)$ rectangle about the x' and y' axes passing through its centroid.

Indicate that the result is a *centroidal* moment of inertia by putting a bar over the symbol I .

Solution. We can use the same approach with $dA = dy dx$, but now the limits of integration over y are now from $-h/2$ to $h/2$.

$$\begin{aligned}
 \bar{I}_{x'} &= \int_A y^2 dA \\
 &= \int_0^b \int_{-h/2}^{h/2} y^2 dy dx
 \end{aligned}$$

$$\begin{aligned}
 &= \int_0^b \left[\frac{y^3}{3} dy \right]_{-h/2}^{h/2} dx \\
 &= \frac{h^3}{12} \int_0^b dx \\
 \bar{I}_{x'} &= \frac{bh^3}{12}
 \end{aligned}$$

Notice that the centroidal moment of inertia of the rectangle is smaller than the corresponding moment of inertia about the baseline. The solution for $\bar{I}_{y'}$ is similar.

Thinking Deeper 10.2.5 Stresses in a Rectangular Beam.

To provide some context for area moments of inertia, let's examine the internal forces in a elastic beam. Assume that some external load is causing an external bending moment which is opposed by the internal forces exposed at a cut.

When an elastic beam is loaded from above, it will sag. Fibers on the top surface will compress and fibers on the bottom surface will stretch, while somewhere in between the fibers will neither stretch or compress. The points where the fibers are not deformed defines a transverse axis, called the **neutral axis**. The neutral axis passes through the centroid of the beam's cross section.

The change in length of the fibers are caused by internal compression and tension forces which increase linearly with distance from the neutral axis. The internal forces sum to zero in the horizontal direction, but they produce a net couple-moment which resists the external bending moment.

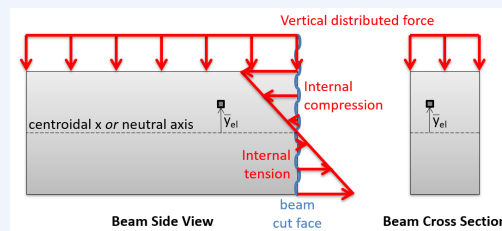


Figure 10.2.6 Internal forces in a beam caused by an external load.

Think about summing the internal moments about the neutral axis on the beam cut face. This moment at a point on the face increases with the square of the distance y of the point from the neutral axis because both the internal force and the moment arm are proportional to this distance. The appearance of y^2 in this relationship is what connects a bending beam to the area moment of inertia.

The shape of the beam's cross-section determines how easily the beam

bends. A beam with more material farther from the neutral axis will have a larger moment of inertia and be stiffer. Of course, the material of which the beam is made is also a factor, but it is independent of this geometrical factor.

10.2.2 Moment of Inertia of a Triangle

We saw in the last section that when solving (10.1.3) the double integration could be conducted in either order, and that the result of completing the inside integral was a single integral. We will use these observations to optimize the process of finding moments of inertia for other shapes by avoiding double integration.

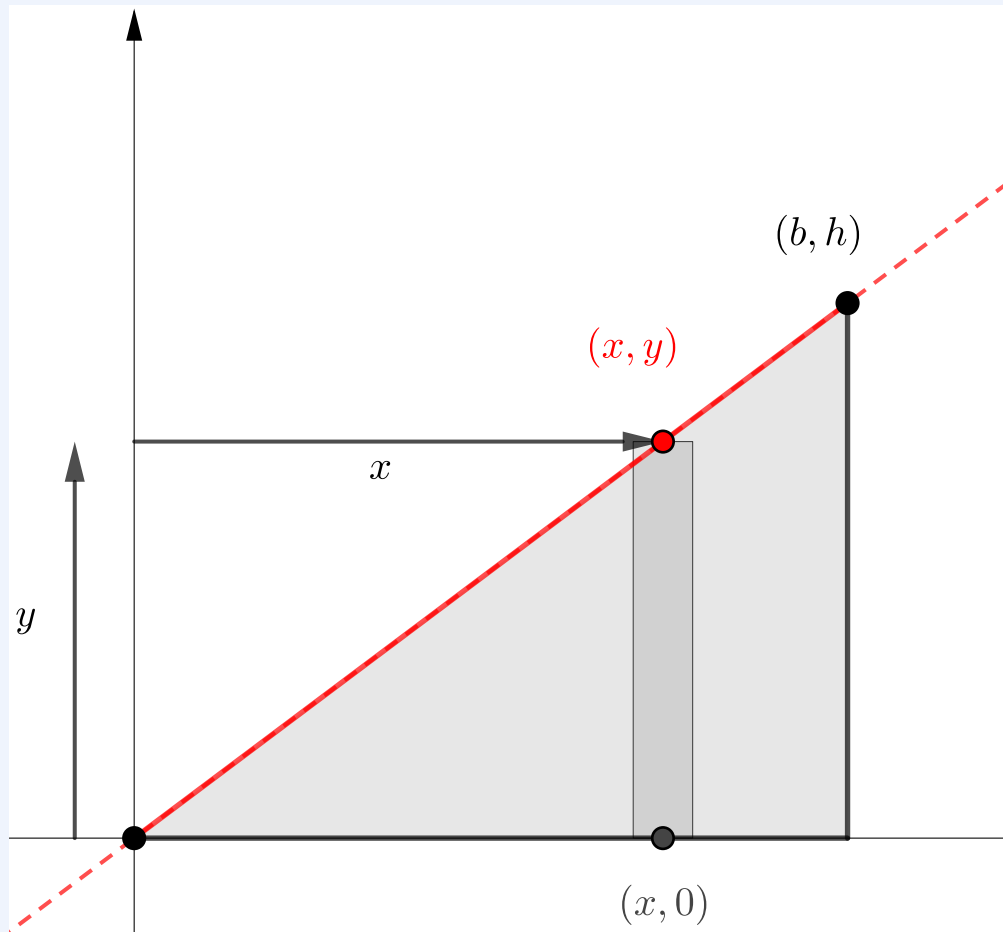
The most straightforward approach is to use the definitions of the moment of inertia (10.1.3) along with strips *parallel* to the designated axis, i.e. horizontal strips when you want to find the moment of inertia about the x axis and vertical strips for the moment of inertia about the y axis.

The strip must be parallel in order for (10.1.3) to work; when parallel, all parts of the strip are the same distance from the axis.

This approach only works if the bounding function can be described as a function of y and as a function of x , to enable integration with respect to x for the vertical strip, and with respect to y for the horizontal strip.

Example 10.2.7 Triangle.

Find the moment of inertia of the $(b \times h)$ right triangle with respect to the x and y axes?



[Live Link](#)

Solution. As we did when finding centroids in [Section 7.7](#) we need to evaluate the bounding function of the triangle. The bottom are constant values, $y = 0$ and $x = b$, but the top boundary is a straight line passing through the origin and the point at (b, h) , which has the equation

$$y(x) = \frac{h}{b}x. \quad (10.2.4)$$

By inspection we see that the a vertical strip extends from the x axis to the function so $dA = y \, dx$.

Since vertical strips are parallel to the y axis we can find I_y by evaluating this integral with $dA = y \, dx$, and substituting $\frac{h}{b}x$ for y

$$I_y = \int_A x^2 \, dA$$

$$\begin{aligned}
&= \int_0^b x^2 y \, dx \\
&= \int_0^b x^2 \left(\frac{h}{b} x \right) dx \\
&= \frac{h}{b} \int_0^b x^3 dx \\
&= \frac{h}{b} \frac{x^4}{4} \Big|_0^b \\
I_y &= \frac{hb^3}{4}
\end{aligned}$$

Similarly we will find I_x using horizontal strips, by evaluating this integral with $dA = (b - x)dy$

$$I_x = \int_A y^2 dA$$

We are expressing dA in terms of dy , so everything inside the integral must be constant or expressed in terms of y in order to integrate. In particular, we will need to solve (10.2.4) for x as a function of y . This is not difficult.

$$x(y) = \frac{b}{h}y.$$

Once this has been done, evaluating the integral is straightforward.

$$\begin{aligned}
I_x &= \int_A y^2 \, dA \\
&= \int_0^h y^2 (b - x) \, dy \\
&= \int_0^h y^2 \left(b - \frac{b}{h}y \right) dy \\
&= b \int_0^h y^2 dy - \frac{b}{h} \int_0^h y^3 dy \\
&= \frac{bh^3}{3} - \frac{b}{h} \frac{h^4}{4} \\
I_x &= \frac{bh^3}{12}
\end{aligned}$$

This is the moment of inertia of a right triangle about an axis passing through its base. By reversing the roles of b and h , we also now have the moment of inertia of a right triangle about an axis passing through its vertical side.

$$I_y = \frac{hb^3}{12}.$$

10.2.3 Moment of Inertia of a Differential Strip

We saw in [Subsection 10.2.2](#) that a straightforward way to find the moment of inertia using a single integration is to use strips which are parallel to the axis of interest, so use vertical strips to find I_y and horizontal strips to find I_x .

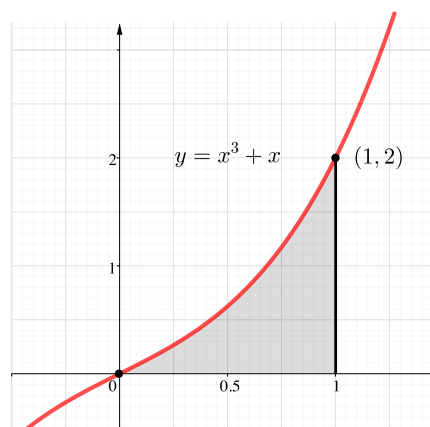
This method requires expressing the bounding function both as a function of x and as a function of y : $y = f(x)$ and $x = g(y)$. There are many functions where converting from one form to the other is not easy.

As an example, let's try finding I_x and I_y for the spandrel bounded by

$$y = f(x) = x^3 + x, \text{ the } x \text{ axis, and } x = 1.$$

Finding I_y using vertical strips is relatively easy. Letting $dA = y \, dx$ and substituting $y = f(x) = x^3 + x$ we have

$$\begin{aligned} I_y &= \int_A x^2 \, dA \\ &= \int_0^1 x^2 y \, dx \\ &= \int_0^1 x^2 (x^3 + x) \, dx \\ &= \int_0^1 (x^5 + x^3) \, dx \\ &= \left. \frac{x^6}{6} + \frac{x^4}{4} \right|_0^1 \\ I_y &= \frac{5}{12} \end{aligned}$$



Finding I_x using horizontal strips is anything but easy. In fact, the integral that needs to be solved is this monstrosity

$$\begin{aligned} I_x &= \int_A y^2 (1 - x) \, dy \\ &= \int_0^2 y^2 \left(1 - \frac{\sqrt[3]{2} \left(\sqrt{81y^2 + 12} + 9y \right)^{2/3} - 2\sqrt[3]{3}}{6^{2/3} \sqrt[3]{\sqrt{81y^2 + 12} + 9y}} \right) dy \end{aligned}$$

... and then a miracle occurs

$$I_x = \frac{49}{120}$$

Clearly, a better approach would be helpful.

When using strips which are *parallel* to the axis of interest is impractical mathematically, the alternative is to use strips which are *perpendicular* to the axis.

Applying our previous result (10.2.2) to a vertical strip with height h and infinitesimal width dx gives the strip's differential moment of inertia. In most cases, h will be a function of x .

$$I_x = \frac{bh^3}{3} \quad \rightarrow \quad dI_x = \frac{h^3}{3} dx \quad (10.2.5)$$

. This is the same result that we saw previously (10.2.3) after integrating the inside integral for the moment of inertia of a rectangle.

This result makes it much easier to find I_x for the spandrel that was nearly impossible to find with horizontal strips.

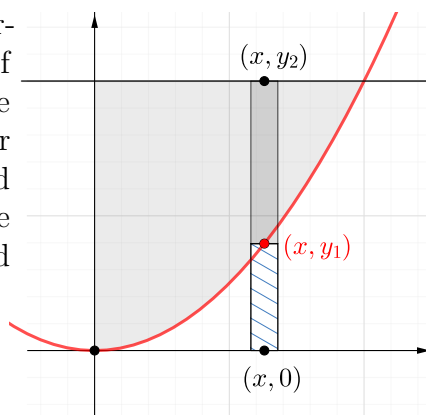
$$\begin{aligned} I_x &= \int_A dI_x = \int_0^1 \frac{y^3}{3} dx \\ &= \int_0^1 \frac{(x^3 + x)^3}{3} dx \\ &= \frac{1}{3} \int_0^1 (x^9 + 3x^7 + 3x^5 + x^3) dx \\ &= \frac{1}{3} \left[\frac{x^{10}}{10} + \frac{3x^8}{8} + \frac{3x^6}{6} + \frac{x^4}{4} \right]_0^1 \\ &= \frac{1}{3} \left[\frac{1}{10} + \frac{3}{8} + \frac{3}{6} + \frac{1}{4} \right] \\ &= \frac{1}{3} \left[\frac{12 + 45 + 60 + 30}{120} \right] \\ I_x &= \frac{49}{120} \end{aligned}$$

The same approach can be used with a horizontal strip dy high and b wide, in which case we have

$$I_y = \frac{b^3 h}{3} \quad \rightarrow \quad dI_y = \frac{b^3}{3} dy \quad (10.2.6)$$

. The width b will usually have to be expressed as a function of y .

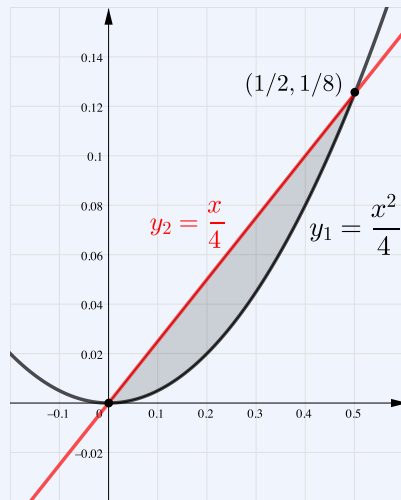
The expression for dI_x assumes that the vertical strip has a lower bound on the x axis. If this is not the case, then find the dI_x for the area between the bounds by subtracting dI_x for the rectangular element below the lower bound from dI_x for the element from the x axis to the upper bound. A similar procedure can be used for horizontal strips.



$$dI_x = \frac{y_2^3}{3} - \frac{y_1^3}{3} = \frac{1}{3}(y_2^3 - y_1^3)$$

This approach is illustrated in the next example.

Example 10.2.8 Moment of Inertia for Area Between Two Curves.



Use vertical strips to find both I_x and I_y for the area bounded by the functions

$$y_1 = x^2/2 \text{ and,}$$

$$y_2 = x/4$$

. Units are cm.

Solution.

1. Set up the integral.

The area is bounded by the functions

$$y_2 = x/4$$

$$y_1 = x^2/2$$

By equating the two functions, we learn that they intersect at $(0,0)$ and $(1/2, 1/8)$, so the limits on x are $x = 0$ and $x = 1/2$.

The differential area dA for vertical strip is

$$dA = (y_2 - y_1) dx = \left(\frac{x}{4} - \frac{x^2}{2} \right) dx.$$

2. Find I_y .

For vertical strips, which are parallel to the y axis we can use the definition of the Moment of Inertia.

$$\begin{aligned} I_y &= \int x^2 dA \\ &= \int_0^{0.5} x^2 \left(\frac{x}{4} - \frac{x^2}{2} \right) dx \\ &= \int_0^{1/2} \left(\frac{x^3}{4} - \frac{x^4}{2} \right) dx \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{x^4}{16} - \frac{x^5}{12} \right) \Big|_0^{1/2} \\
&= \left(\frac{(1/2)^4}{16} - \frac{(1/2)^5}{10} \right) \\
&= \frac{1}{64} \left(\frac{1}{4} - \frac{1}{5} \right) \\
I_y &= \frac{1}{1280} = 7.81 \times 10^{-4} \text{ cm}^4
\end{aligned}$$

3. Find I_x .

For vertical strips, which are perpendicular to the x axis, we will subtract the moment of inertia of the area below y_1 from the moment of inertia of the area below y_2 .

$$\begin{aligned}
I_x &= \int_{A_2} dI_x - \int_{A_1} dI_x \\
&= \int_0^{1/2} \frac{y_2^3}{3} dx - \int_0^{1/2} \frac{y_1^3}{3} dx \\
&= \frac{1}{3} \int_0^{1/2} \left[\left(\frac{x}{4} \right)^3 - \left(\frac{x^2}{2} \right)^3 \right] dx \\
&= \frac{1}{3} \int_0^{1/2} \left[\frac{x^3}{64} - \frac{x^6}{8} \right] dx \\
&= \frac{1}{3} \left[\frac{x^4}{256} - \frac{x^7}{56} \right]_0^{1/2} \\
I_x &= \frac{1}{28672} = 3.49 \times 10^{-6} \text{ cm}^4
\end{aligned}$$

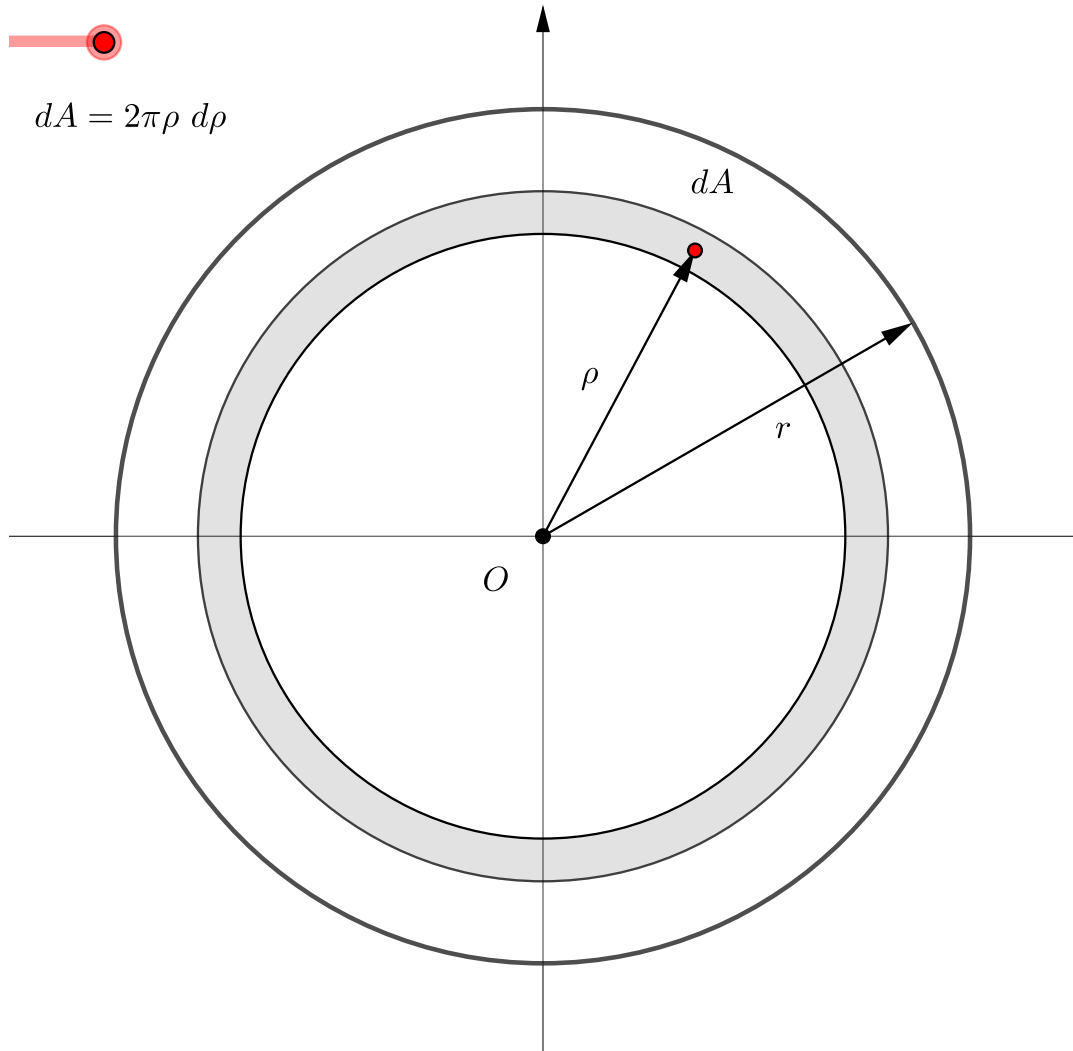
10.2.4 Circles, Semicircles, and Quarter-circles

In this section, we will use polar coordinates and symmetry to find the moments of inertia of circles, semi-circles and quarter-circles.

We will start by finding the polar moment of inertia of a circle with radius r , centered at the origin. You will recall from [Subsection 10.1.4](#) that the polar moment of inertia is similar to the ordinary moment of inertia, except the distance squared term is the distance from the element to a point in the plane rather than the perpendicular distance to an axis, and it uses the symbol J with a subscript indicating the point.

To take advantage of the geometry of a circle, we'll divide the area into thin rings, as shown in the diagram, and define the distance from the origin to a point on the ring as ρ . The reason for using thin rings for dA is the same reason

we used strips parallel to the axis of interest to find I_x and I_y ; all points on the differential ring are the same distance from the origin, so we can find the moment of inertia using single integration.



[Live Link](#)

Figure 10.2.9 Circle divided into differential rings

The differential area of a circular ring is the circumference of a circle of radius ρ times the thickness $d\rho$.

$$dA = 2\pi\rho d\rho.$$

Adapting the basic formula for the polar moment of inertia (10.1.5) to our labels, and noting that limits of integration are from $\rho = 0$ to $\rho = r$, we get

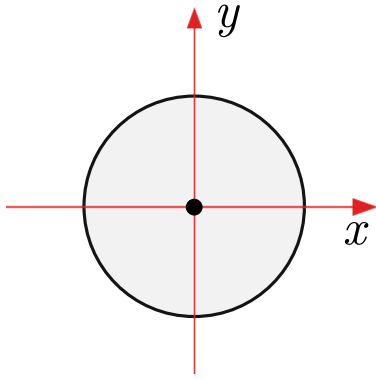
$$J_O = \int_A r^2 dA \quad \rightarrow \quad J_O = \int_0^r \rho^2 2\pi\rho d\rho \quad (10.2.7)$$

Proceeding with the integration,

$$\begin{aligned}
 J_O &= \int_0^r \rho^2 2\pi\rho \, d\rho \\
 &= 2\pi \int_0^r \rho^3 d\rho \\
 &= 2\pi \left[\frac{\rho^4}{4} \right]_0^r \\
 J_O &= \frac{\pi r^4}{2}
 \end{aligned} \tag{10.2.8}$$

This is the polar moment of inertia of a circle about a point at its center.

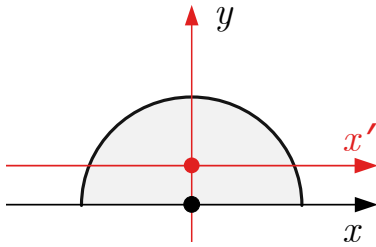
With this result, we can find the rectangular moments of inertia of circles, semi-circles and quarter circle simply. Noting that the polar moment of inertia of a shape is the sum of its rectangular moments of inertia I_x and I_y , these are equal to each other for a circle due to its symmetry. Therefore, by (10.5.2), which is easily proven,



$$\begin{aligned}
 J_O &= I_x + I_y \\
 \bar{I}_x = \bar{I}_y &= \frac{J_O}{2} = \frac{\pi r^4}{4}
 \end{aligned} \tag{10.2.9}$$

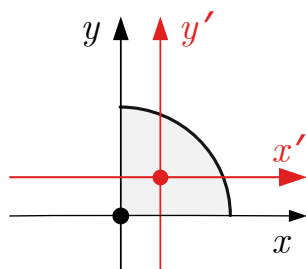
This is the moment of inertia of a circle about a vertical or horizontal axis passing through its center.

A circle consists of two semi-circles above and below the x axis, so the moment of inertia of a semi-circle about a diameter on the x axis is just half of the moment of inertia of a whole circle. The moment of inertia about the vertical centerline is the same.



$$I_x = \bar{I}_y = \frac{\pi r^4}{8}. \tag{10.2.10}$$

Similarly, the moment of inertia of a quarter circle is half the moment of inertia of a semi-circle, so



$$I_x = I_y = \frac{\pi r^4}{16}. \quad (10.2.11)$$

In these diagrams, the centroidal axes are red, and moments of inertia about centroidal axes are indicated by the overbar. We will see how to use the parallel axis theorem to find the centroidal moments of inertia for semi- and quarter-circles in [Section 10.3](#).

10.2.5 Summary of Integration Techniques

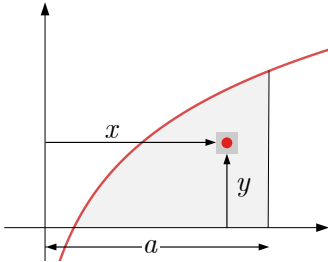
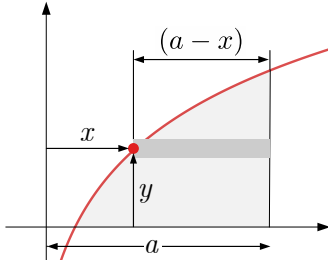
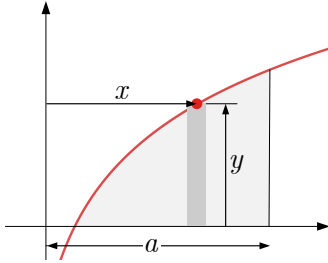
Here is a summary of the alternate approaches to finding the moment of inertia of a shape using integration.

You may choose to divide the shape into square differential elements to compute the moment of inertia using the fundamental definitions. The disadvantage of this approach is that you need to set up and compute a double integral. Identifying the correct limits on the integrals is often difficult.

If you would like to avoid double integration, you may use vertical or horizontal strips, but you must take care to apply the correct integral. If you use vertical strips to find I_y or horizontal strips to find I_x , then you can still use [\(10.1.3\)](#), but skip the double integration. When the entire strip is the same distance from the designated axis, integrating with a parallel strip is equivalent to performing the inside integration of [\(10.1.3\)](#).

As we have seen, it can be difficult to solve the bounding functions properly in terms of x or y to use parallel strips. In this case, you can use vertical strips to find I_x or horizontal strips to find I_y by integrating the differential moment of inertia of the strip as discussed in [Subsection 10.2.3](#).

Table 10.2.10 Moment of Inertia Integration Strategies

Element	dA	dI
	$dA = dx \, dy$ or $dA = dy \, dx$	$dI_x = y^2 \, dA$ $dI_y = x^2 \, dA$
	$dA = (a - x) \, dy$	$dI_x = y^2 \, dA$ $dI_y = \frac{(a^3 - x^3)}{3} \, dx$
	$dA = y \, dx$	$dI_x = \frac{y^3}{3} \, dx$ $dI_y = x^2 \, dA$

10.3 Parallel Axis Theorem

The parallel axis theorem relates the moment of inertia of a shape about an arbitrary axis to its moment of inertia about a parallel centroidal axis.

This theorem is particularly useful because if we know the centroidal moment of inertia of a shape, we can calculate its moment of inertia about any parallel axis by adding an appropriate correction factor. Alternately, if we know the moment of inertia about an axis, we can find the associated centroidal moment of inertia by subtracting the same factor.

The centroidal moment of inertia of common shapes are well known, and readily available in tables of properties of shapes such as [Subsection 10.3.2](#).

10.3.1 Derivation

We will use the defining equation for the moment of inertia (10.1.3) to derive the parallel axis theorem.

The diagram shows an arbitrary shape, and two parallel axes: the x' axis, drawn in red, passes through the centroid of the shape at C , and the x axis, which is parallel and separated by a distance, d . The shape has area A , which is divided into square differential elements dA . The distance from the x axis to the element dA is y , and the distance from the x' axis is y' .

By (10.1.3), the moment of inertia of the shape about the x and x' axes are

$$I_x = \int_A y^2 dA \quad \bar{I}_{x'} = \int_A (y')^2 dA$$

The first is the value we are looking for, and the second is the centroidal moment of inertia of the shape. These two are related through the distance d , because $y = d + y'$. Substituting that relation into the first equation and expanding the binomial gives

$$\begin{aligned} I_x &= \int_A (d + y')^2 dA \\ &= \int_A [(y')^2 + 2 y' d + d^2] dA \\ &= \int_A (y')^2 dA + 2d \int_A y' dA + d^2 \int_A dA \end{aligned}$$

You should recognize these three integrals. The first is the centroidal moment of inertia of the shape $\bar{I}_{x'}$, and the third is the total area of the shape, A . The middle integral is $Q_{x'}$, the [first moment of area](#) (10.1.2) with respect to the centroidal axis x' . So we have,

$$I_x = \bar{I}_{x'} + 2dQ_{x'} + d^2A.$$

Furthermore, $Q_{x'}$ is exactly zero because the x' axis passes through the centroid, meaning that elements of area above and below the centroidal axis exactly balance and cancel each other out. After dropping the middle term we get the version of the parallel axis theorem which you should remember,

$$I = \bar{I} + Ad^2. \quad (10.3.1)$$

The subscripts designating the x and x' axes have been dropped because this equation is applicable to any direction of parallel axes, not specifically horizontal axes.

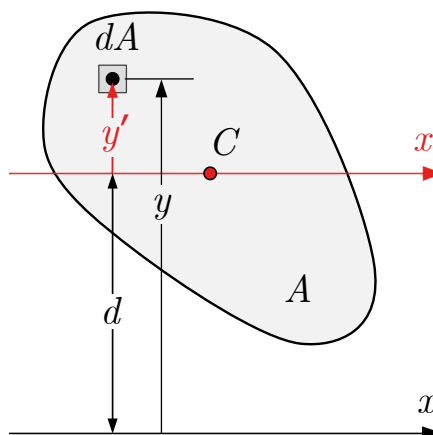
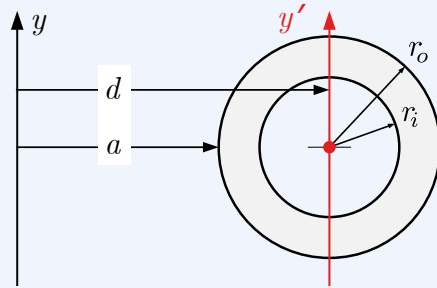


Figure 10.3.1 Definitions for the parallel axis theorem.

This equation says that you find the moment of inertia of a shape about *any* axis by adding Ad^2 to the *parallel* centroidal moment of inertia. You can consider the Ad^2 term as ‘correction factor’ to account for the distance of the axis from the centroid. This term is always positive, so the centroidal moment of inertia is always the minimum moment of inertia for a particular axis direction.

The next example show how the parallel axis theorem is typically used to find the moment of inertia of a shape about an axis, by using then centroidal moment of inertia formulas found in [Subsection 10.3.2](#).

Example 10.3.2 Circular Ring.



Use the parallel axis theorem to find the moment of inertia of the circular ring about the y axis.

The dimensions of the ring are $R_i = 30$ mm, $R_o = 45$ mm, and $a = 80$ mm.

Solution. To apply the parallel axis theorem, we need three pieces of information

1. The centroidal moment of inertia of the ring, I_y ,
2. the area of the ring, A ,
3. the distance between the two parallel axes, d .

The area of the ring is found by subtracting the area of the inner circle from the area of the outer circle. The centroidal moment of inertia is calculated similarly using [\(10.2.9\)](#). The distance between the y and y' axis is available from the diagram. Inserting these values into the parallel axis theorem gives,

$$\begin{aligned}
 I_y &= I_y + Ad^2 \\
 &= \underbrace{\frac{\pi}{4}(r_o^4 - r_i^4)}_{I_y} + \underbrace{\pi(r_o^2 - r_i^2)}_A \underbrace{(a + r_o)^2}_{d^2} \\
 &= \frac{\pi}{4}(45^4 - 30^4) + \pi(45^2 - 30^2)(80 + 45)^2 \\
 &= 2.58 \times 10^6 \text{ mm}^4 + 55.2 \times 10^6 \text{ mm}^4 \\
 I_y &= 57.8 \times 10^6 \text{ mm}^4
 \end{aligned}$$

It is interesting that the ‘correction factor’ is more than 20 times greater than the centroidal moment of inertia of the ring. This indicates the

importance of the distance squared term on the moment of inertia of a shape.

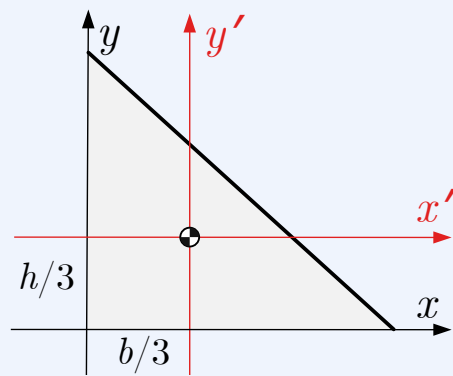
You may feel like the answer to this problem is “too big”. Large answers are normal in problems like this because the moment of inertia involves raising lengths the fourth power.

If it really bothers you, you can convert the results from mm^4 to m^4 , but then the number will probably feel “too small” to you. It’s best not to worry about it.

The parallel axis theorem can also be used to find a centroidal moment of inertia when you already know the moment of inertia of a shape about another axis, by using the theorem ‘backwards’,

$$I = \bar{I} + Ad^2 \quad \rightarrow \quad \bar{I} = I - Ad^2$$

Example 10.3.3 Centroidal Moment of Inertia of a Triangle.



Find the centroidal moment of inertia of a triangle knowing that the moment of inertia about its base is

$$I_x = \frac{1}{12}bh^3.$$

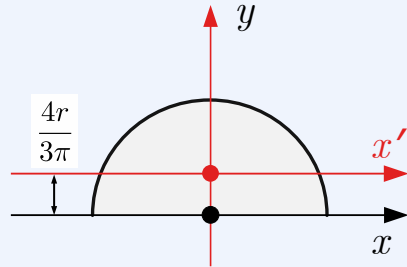
Solution. For the triangle we have the following information: $I_x = bh^3/12$, $A = bh/2$, and $d = \bar{y} = h/3$.

Looking for $\bar{I}_{x'}$:

$I = \bar{I} + Ad^2$	Parallel Axis Theorem, general statement
$\bar{I}_{x'} = I_x - A(\bar{y})^2$	Parallel Axis Theorem, backwards
$= \frac{bh^3}{12} - \frac{bh}{2} \left(\frac{h}{3}\right)^2$	Insert values for triangle
$= bh^3 \left[\frac{1}{12} - \frac{1}{18} \right]$	Factor out common terms
$= bh^3 \left[\frac{3}{36} - \frac{2}{36} \right]$	Common denominator
$\bar{I}_{x'} = \frac{bh^3}{36}$	Result

The procedure for $\bar{I}_y$ is similar, or you can simply reverse the roles of b and h .

Example 10.3.4 Centroidal Moment of inertia of a Semi-Circle.



Find the centroidal moment of inertia of a semi-circle knowing that the moment of inertia about its base is

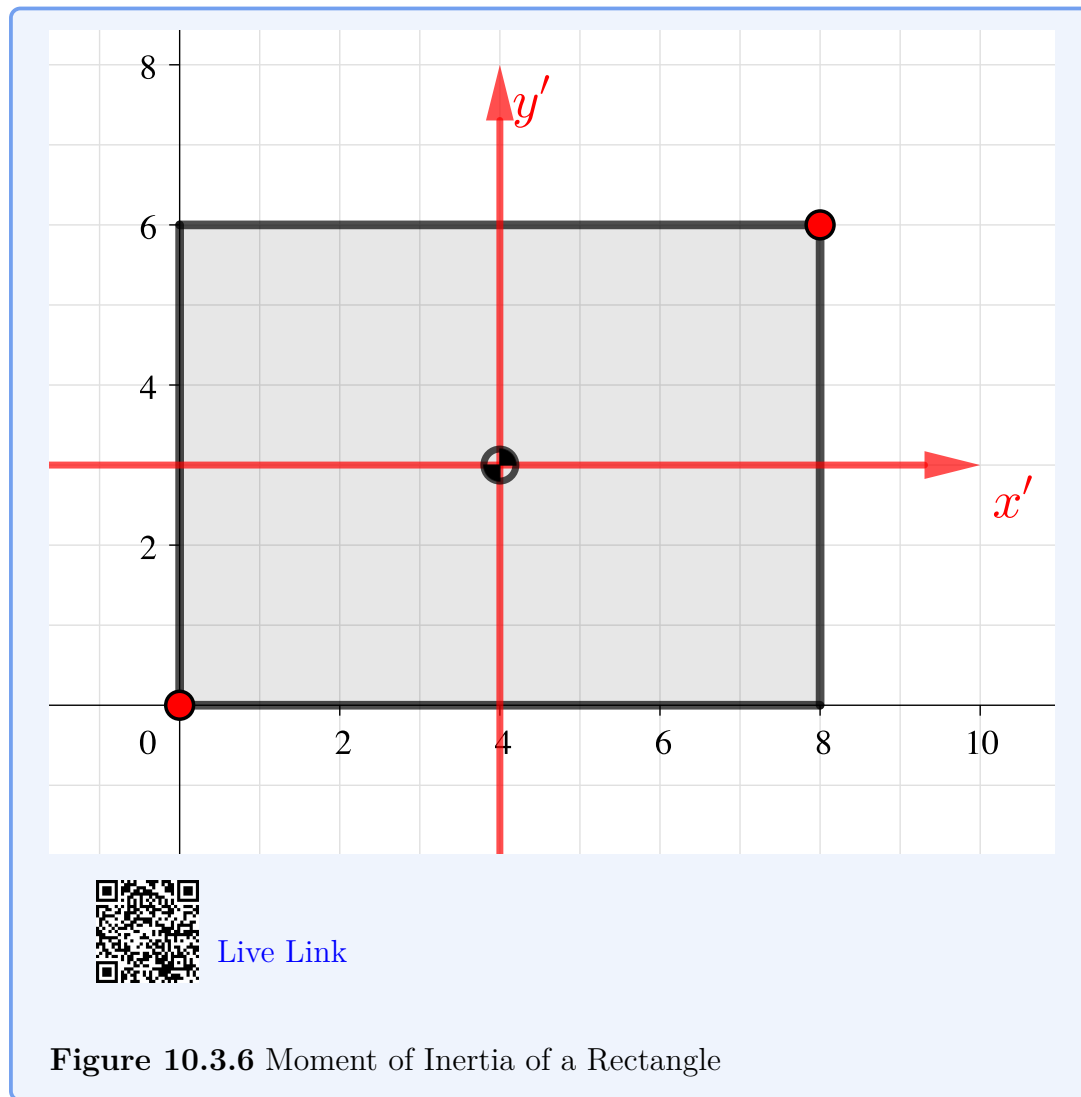
$$I_x = \frac{\pi}{8} r^4.$$

Solution. The area of a semicircle is $A = \frac{\pi r^2}{2}$ and the distance between the parallel axes is $d = \frac{4r}{3\pi}$, so

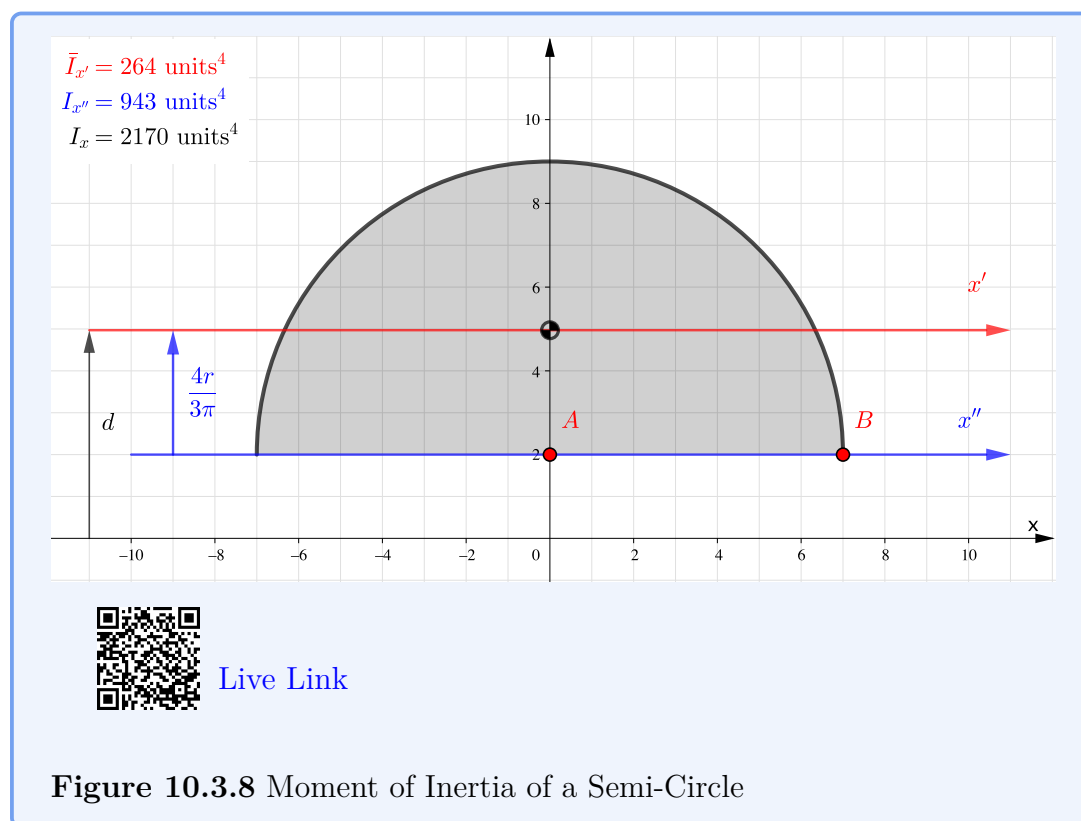
$$\begin{aligned}\bar{I}_{x'} &= I_x - Ad^2 \\ &= \frac{\pi}{8} r^4 - \left(\frac{\pi r^2}{2} \right) \left(\frac{4r}{3\pi} \right)^2 \\ \bar{I}_{x'} &= \left(\frac{\pi}{8} - \frac{8}{9\pi} \right) r^4\end{aligned}$$

Example 10.3.5 Interactive: Rectangle.

This interactive allows you to change the location and size of the grey rectangle. Try to compute both the centroidal area moment of inertia $\bar{I}_{x'}$ and $\bar{I}_{y'}$ and the area moment of inertia about the system axes I_x and I_y

**Example 10.3.7** Interactive: Semi-Circle.

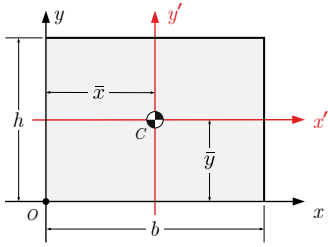
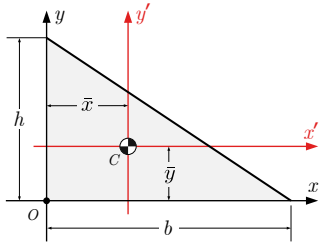
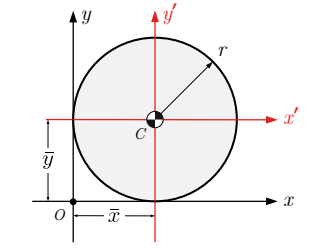
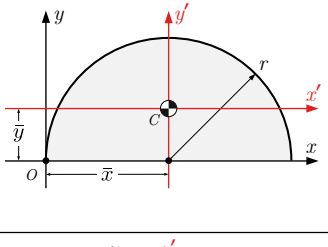
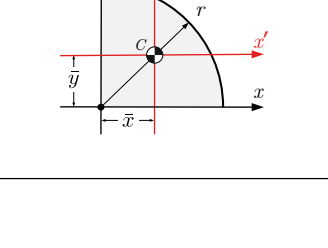
Use this interactive to practice computing the area moments of inertia of the semi-circle about the centroidal x' axis, the bottom edge x'' , and the system x axis. You can change the location and size of the semicircle by moving the red points..



10.3.2 Moments of Inertia Table

This table summarizes the properties of the common shapes discussed previously.

Table 10.3.9 Moments of Inertia of Common Shapes

Shape	Centroid	Centroidal MOI	I_x, I_y
	$(b/2, h/2)$	$\bar{I}_{x'} = \frac{1}{12}bh^3$ $\bar{I}_{y'} = \frac{1}{12}b^3h$	$I_x = \frac{1}{3}bh^3 I_y = \frac{1}{3}b^3h$
	$(b/3, h/3)$	$\bar{I}_{x'} = \frac{1}{36}bh^3 \bar{I}_{y'} = \frac{1}{36}b^3h$	$I_x = \frac{1}{12}bh^3 I_y = \frac{1}{12}b^3h$
	(r, r)	$\bar{I}_{x'} = \bar{I}_{y'} = \frac{\pi}{4}r^4$	$I_x = I_y = \frac{5\pi}{4}r^4$
	$\left(r, \frac{4r}{3\pi}\right)$	$\bar{I}_{x'} = \left(\frac{\pi}{8} - \frac{8}{9\pi}\right)r^4$ $\bar{I}_{x'} \approx 0.1098 r^4$ $\bar{I}_{y'} = \frac{\pi}{8}r^4$	$I_x = \frac{\pi}{8}r^4$ $I_y = \frac{5\pi}{8}r^4$
	$\left(\frac{4r}{3\pi}, \frac{4r}{3\pi}\right)$	$\bar{I}_{x'} = \frac{1}{2} \left(\frac{\pi}{8} - \frac{8}{9\pi}\right)r^4$ $\bar{I}_{x'} = \bar{I}_{y'} \approx 0.0549 r^4$	$I_x = I_y = \frac{\pi}{16}r^4$

10.4 Composite Shapes

Key Questions

- Where do the common shape area moment of inertia equations come from?
- What is the parallel axis theorem?
- When do you need to apply the parallel axis theorem?
- About which point do you find the smallest area moments of inertia? What is it about this point that is so special?

In this section we will find the moment of inertia of shapes formed by combining simple shapes like rectangles, triangles and circles much the same way we did to find centroids in [Section 7.5](#).

The procedure is to divide the complex shape into its sub shapes and then use the centroidal moment of inertia formulas from [Subsection 10.3.2](#), along with the parallel axis theorem ([10.3.1](#)) to calculate the moments of inertia of parts, and finally combine them to find the moment of inertia of the original shape.

10.4.1 Composite Area Method

For a composite shape made up of n subparts, the moment of inertia of the whole shape is the sum of the moments of inertia of the individual parts, however the moment of inertia of any holes are subtracted from the total of the positive areas.

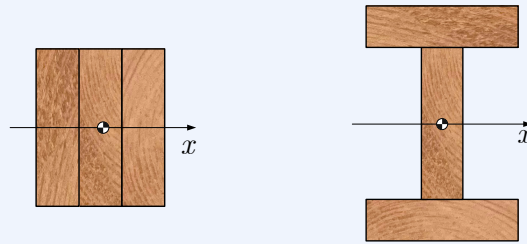
Moments of inertia are always calculated relative to a specific axis, so the moments of inertia of all the sub shapes must be calculated with respect to this same axis, which will usually involve applying the parallel axis theorem.

$$I = \sum_{i=0}^n (I)_i = \sum_{i=0}^n (\bar{I} + Ad^2)_i \quad (10.4.1)$$

The method is demonstrated in the following examples.

Example 10.4.1 Beam Design.

You have three 24 ft long wooden 2×6's and you want to nail them together them to make the stiffest possible beam. The stiffness of a beam is proportional to the moment of inertia of the beam's cross-section about a horizontal axis passing through its centroid. The actual dimensions of nominal 2×6 lumber are 1.5 in by 5.5 in.



Which of the arrangements will be the stiffest, and what is the ratio of the two moments of inertia?

Solution. Given: $b = 1.5$ in, $h = 5.5$ in.

In case 1 the centroids of all three rectangles are on the x axis, so the parallel axis theorem is unnecessary.

$$\begin{aligned}
 (I_x)_1 &= \sum_{i=1}^3 \bar{I} + Ad^2 \\
 &= 3 \frac{bh^3}{12} \\
 &= \frac{(1.5)(5.5)^3}{4} \\
 (I_x)_1 &= 62.4 \text{ in}^4
 \end{aligned}$$

This value is the same as the moment of inertia of a $(4.5 \text{ in} \times 5.5 \text{ in})$ rectangle about its centroid.

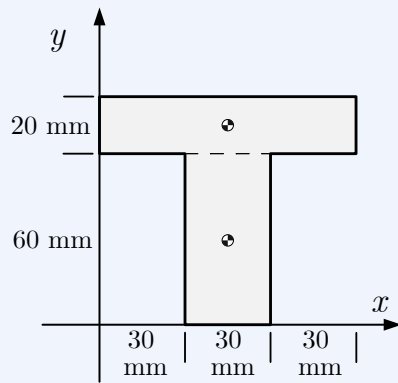
In case 2, the parallel axis theorem must be used for the upper and lower rectangles, since their centroids are not on the x axis.

$$\begin{aligned}
 (I_x)_2 &= \sum_{i=1}^3 \bar{I} + Ad^2 \\
 &= \left(\frac{bh^3}{12} \right) + 2 \left[\frac{1}{12} hb^3 + (bh)(h/2 + b/2)^2 \right] \\
 &= \frac{(1.5)(5.5)^3}{12} + 2 \left[\frac{(1.5)^3(5.5)}{12} + (1.5 \times 5.5)(3.5)^2 \right] \\
 &= 20.8 \text{ in}^4 + 2 [1.547 \text{ in}^4 + 101.6 \text{ in}^4] \\
 (I_x)_2 &= 226 \text{ in}^4
 \end{aligned}$$

$$(I_x)_2 / (I_x)_1 = 3.61$$

The I-beam is about 3.6 times stiffer than the sandwich beam. This optimization of material usage is the reason we use I-beams.

Example 10.4.2 T Shape.



Find the moment of inertia of the T shape about the x and y axes.

Solution.1. *Strategy.*

Divide the T shape into a 30 mm $\times$ 60 mm vertical rectangle (1), and a 90 mm $\times$ 20 mm horizontal rectangle (2) then add the moments of inertia of the two parts.

$$I_x = (I_x)_1 + (I_x)_2 \qquad I_y = (I_y)_1 + (I_y)_2$$

2. *MOI about the x Axis.*

The bottom edge of rectangle 1 is on the x axis. Using the formula from [Subsection 10.3.2](#) gives

$$(I_x)_1 = \frac{bh^3}{3} = \frac{(30)(60)^3}{3} = 2.16 \times 10^6 \text{ mm}^4.$$

The centroid of rectangle 2 is located 70 mm above the x axis so we must use the parallel axis theorem ([10.3.1](#)), so

$$\begin{aligned} (I_x)_2 &= \bar{I} + Ad^2 \\ &= \frac{bh^3}{12} + (bh)d^2 \\ &= \frac{(90)(20)^3}{12} + (90 \times 20)(70)^2 \\ (I_x)_2 &= 8.88 \times 10^6 \text{ mm}^4 \end{aligned}$$

.

The moment of inertia of the entire T shape about the x axis is the sum of these two values,

$$I_x = (I_x)_1 + (I_x)_2 = 11.04 \times 10^6 \text{ mm}^4.$$

3. MOI about the y Axis.

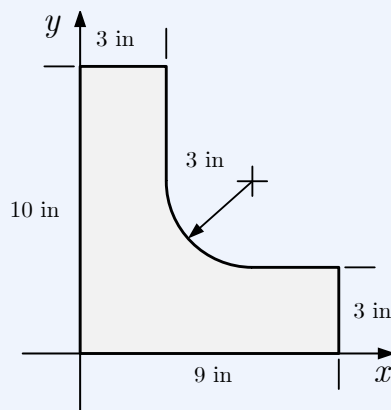
We can use the same procedure to find the moment of inertia about the y axis, however it is usually more convenient to organize all the necessary information in a table rather than writing the equations explicitly.

We will use the parallel axis theorem for both rectangles with d representing the distance between the y axis and the centroid of the part. In this example d is the same for both parts, but that will not always be true.

Part	Dimensions	$\bar{I}_y = hb^3/12$	$A = bh$	d	Ad^2	$I_y = \bar{I} + Ad^2$
Units	mm	mm ⁴	mm ²	mm	mm ⁴	mm ⁴
1	$b = 30, h = 60$	135×10^3	1800	45	3.645×10^6	3.78×10^6
2	$b = 90, h = 20$	1.215×10^6	1800	45	3.645×10^6	4.86×10^6
Total						8.64×10^6

$$I_y = (I_y)_1 + (I_y)_2 = 8.64 \times 10^6 \text{ mm}^4$$

Example 10.4.3 Fillet.



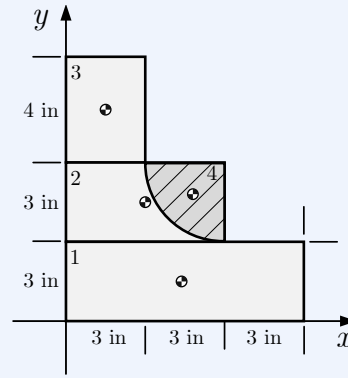
Find the moment inertia of the area about the x axis.

Solution.

1. Strategy.

First, divide the area into four parts:

- (a) a 9 in $\times$ 3 in rectangle
- (b) a 6 in $\times$ 3 in rectangle
- (c) a 3 in $\times$ 4 in rectangle, and
- (d) a removed quarter-circle with a 3 in radius.



Then set up a table and apply the parallel axis theorem (10.3.1) as in the previous example. Since the quarter-circle is removed, subtract its moment of inertia from total of the other shapes.

2. MOI about the y Axis.

The centroidal moment of inertia of a quarter-circle, from Subsection 10.3.2 is

$$\begin{aligned} I_x &= \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) r^4 \\ &= 0.0549 r^4 \end{aligned}$$

The distance from the top edge of the quarter-circle down to its centroid is $\frac{4r}{3\pi} = 1.273$ in, so the distance from the x axis to its centroid is

$$d = 6 - 1.27 = 4.727 \text{ in.}$$

Fill out the table of information.

Part	Dimensions	$\bar{I}_x$	A	$d = \bar{y}$	Ad^2	$I_y = \bar{I} + Ad^2$
Units	inch	inch ⁴	inch ²	inch	inch ⁴	inch ⁴
1	$b = 9, h = 3$	$bh^3/12 = 20.25$	27	1.5	60.75	81
2	$b = 6, h = 3$	$bh^3/12 = 13.5$	18	4.5	364.5	378
3	$b = 3, h = 4$	$bh^3/12 = 16$	12	8	768	784
4	$r = 3$	$0.0549 r^4 = 4.45$	7.07	4.727	158.0	162.4

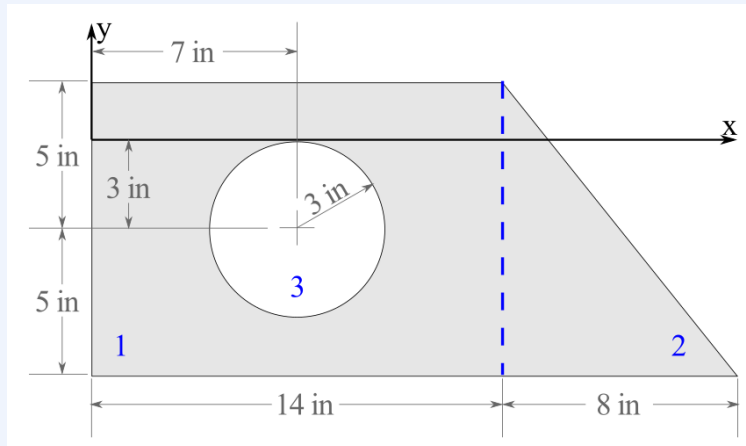
Take care to subtract the moment of inertia of the removed quarter-circle from the total.

$$I_x = (I_x)_1 + (I_x)_2 + (I_x)_3 - 1(I_x)_4 = 1080.6 \text{ in}^4.$$

Example 10.4.4 Concrete Pipe Casing.

The cross section of a concrete pipe casing composed of a rectangular block, a triangular wedge, and a circular pipe formed through the middle of the block is shown below.

As the pipe casing will be subject to various loads, find the area moment of inertia of the cross section about the x and y axes.

**Solution.**1. *Strategy.*

Organize all the necessary information into a table, then total the moments of inertia of the parts to get the moment of inertia of the whole shape. Remember that the hole is removed from the shape, so its contribution to the total moment of inertia is negative.

2. *Table.*

Part Units	A_i in ²	d_{x_i} in	d_{y_i} in	$A_i d_x^2$ in ⁴	$A d_y^2$ in ⁴	$\bar{I}_x$ in ⁴	$\bar{I}_y$ in ⁴
Rectangle 1 $b = 14,$ $h = 10$	bh $= 140$	7	-3	6860	1260	$bh^3/12$ $= 1167$	$b^3h/12$ $= 2287$
Triangle 2 $b = 8,$ $h = 10$	$bh/2$ $= 40$	16.67	-4.67	11111	871.1	$bh^3/36$ $= 222.2$	$b^3h/12$ $= 142.2$
Circular Hole 3 $r = 3$	$-\pi r^2$ $= -28.27$	7	-3	-1385	-254.5	$-\pi r^4/4$ $= -63.62$	$-\pi r^4/4$ $= -63.62$
Total	151.7			16586	1877	1325	2365

3. *Total.*

$$\begin{aligned}
(I_x)_1 &= [\bar{I}_x + Ad_y^2]_1 = 2427 \text{ in}^4 & (I_y)_1 &= [\bar{I}_y + Ad_x^2]_1 = 9147 \text{ in}^4 \\
(I_x)_2 &= [\bar{I}_x + Ad_y^2]_2 = 1093 \text{ in}^4 & (I_y)_2 &= [\bar{I}_y + Ad_x^2]_2 = 11253 \text{ in}^4 \\
(I_x)_3 &= [\bar{I}_x + Ad_y^2]_3 = -318.1 \text{ in}^4 & (I_y)_3 &= [\bar{I}_y + Ad_x^2]_3 = -1449 \text{ in}^4 \\
I_x &= \sum (I_x)_i = 3202 \text{ in}^4 & I_y &= \sum (I_y)_i = 18951 \text{ in}^4
\end{aligned}$$

Alternately, you could find the moments of inertia by adding the sums of the columns, since you are adding the same values together, just in a different order.

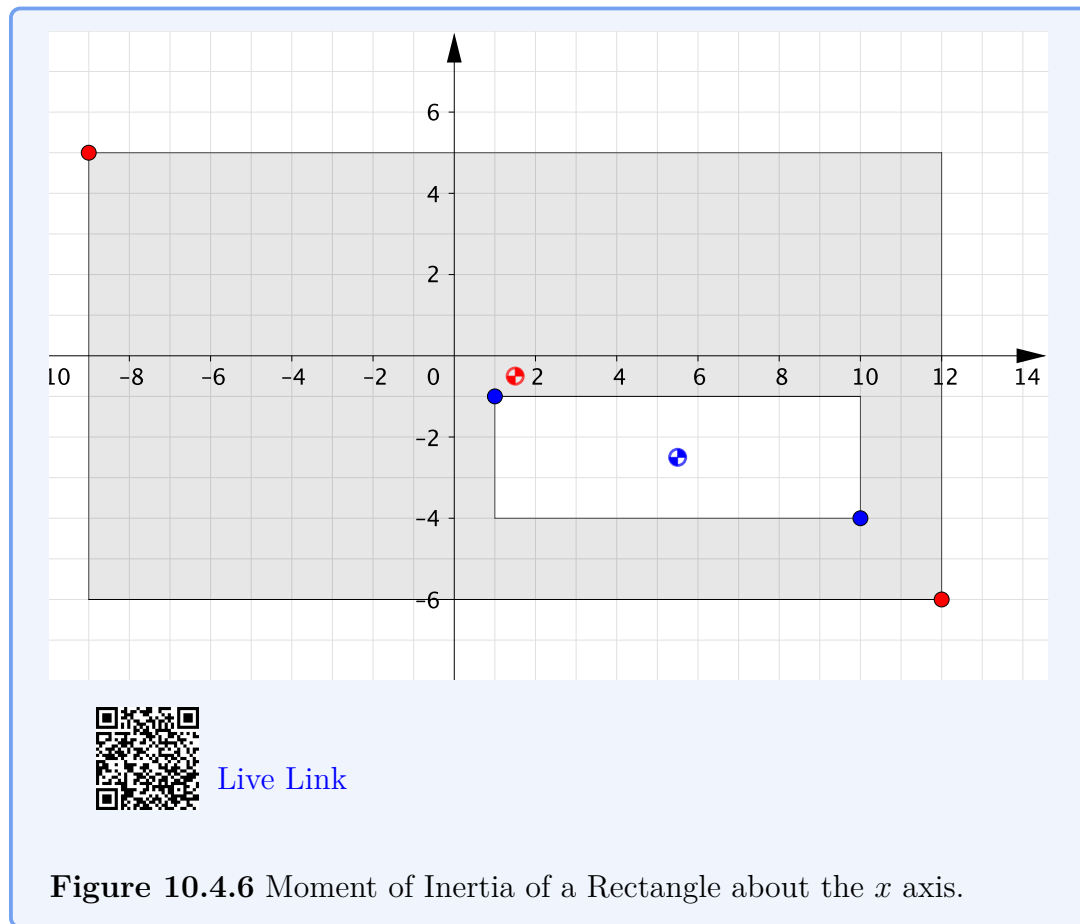
$$I_x = \sum \bar{I}_x + \sum Ad_y^2 = 3202 \text{ in}^4 \quad I_y = \sum \bar{I}_y + \sum Ad_x^2 = 18951 \text{ in}^4$$

Example 10.4.5 Interactive: Composite Rectangles.

This interactive shows a composite shape consisting of a large rectangle with a smaller rectangle subtracted. You can change the location and size of the rectangles by moving the red and blue points.

Use the interactive to see how changes to the rectangles affects the moments of inertia of this shape about the system x axis. Notice that for two-part shapes like this, the centroid of the composite shape is on the line connecting the centroids of the two parts.

For calculations, it is convenient to collect all the needed information in a table as is done here.



10.4.2 Structural Steel Sections

Steel is a strong, versatile, and durable material commonly used for girders, beams, and columns in steel structures such as buildings, bridges, and ships. When possible designers prefer to use prefabricated *Standardized Structural Steel* to minimize material cost.

Structural steel is available in a variety of shapes called *sections*, shown below. These include universal beams and columns (W, S), structural channels (C), equal and unequal angle sections (L), Tee shapes (T), rectangular, square and round hollow structural sections (HSS), bar, rod, and plate. All are available in a range of sizes from small to huge. Steel sections are manufactured by hot or cold rolling or fabricated by welding flat or curved steel plates together.

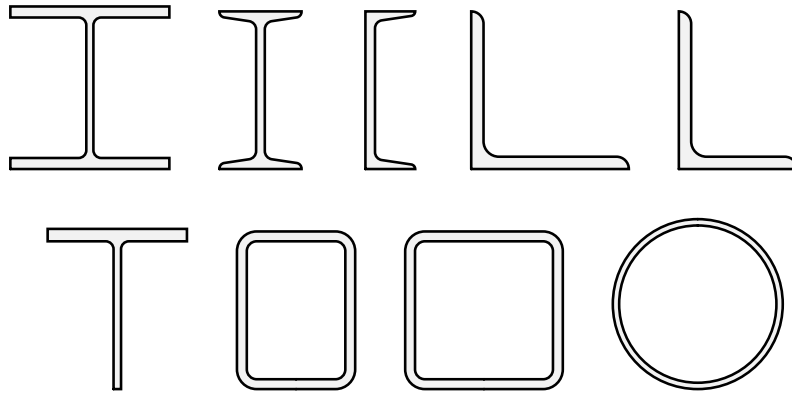


Figure 10.4.7 AISC Standard Sections: Left to right -- Wide-Flange (W), American Standard (S), Channel (C), Equal Angle (L), Unequal Angle (L), Structural Tee (T), Rectangle (HSS), Square (HSS), Round (HSS).

Designers and engineers must select the most appropriate and economical section which can support the potential tension, compression, shear, torsion and bending loads. Tables of properties of Standard Steel Sections are published by the American Institute of Steel Construction, and are used to simplify the process. The tables contain important properties of the sections, including dimensions, cross sectional area, weight per foot, and moment of inertia about vertical and horizontal axes. An abbreviated subset of the AISC tables are available in [Appendix D](#).

In this section we will use the information in the AISC tables to find the moments of inertia of standard sections and also of composite shapes incorporating standard sections.

The top and bottom pieces of an I-beam are called flanges. The middle portion is referred to as the web. The flanges take most of the internal compression and tension forces as they are located the furthest from the neutral axis, and the web mainly acts to support any shear forces and hold the two flanges apart. The transverse axis through the centroid of the cross section is called the neutral axis, and cutting plane through the beam at the neutral axis is called the neutral plane, or neutral surface. This surface does not lengthen or shorten during bending.

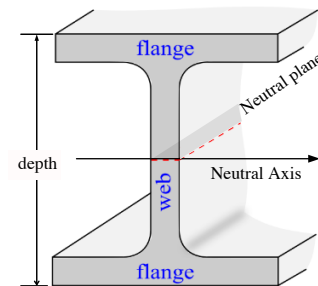
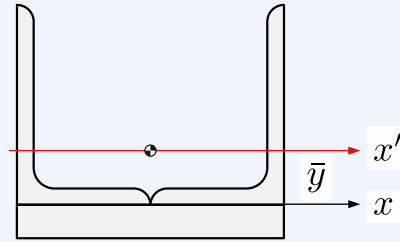


Figure 10.4.8 Beam Nomenclature

Example 10.4.9 Built-up beam.

A built-up beam consists of two L6×4×1/2 angles attached to a 8×1 plate as shown. Determine

- the distance from the x axis to the neutral axis, which passes through the centroid of the combined shape, and
- the moment of inertia of the combined shape about the neutral axis.

**Solution.**1. *Strategy.*

Determine the properties of the sub shapes with respect to the x axis, and then use them to find the neutral axis.

Use the parallel axis theorem to find the moment of inertia of the parts with respect to the neutral axis.

Take advantage of the fact that the two angles are identical and positioned similarly.

2. *Find the neutral axis.*

For one L6 × 4 × 1/2 angle, from [Section D.1](#)

$$A_L = 4.75 \text{ in}^2$$

$$\bar{y}_L = 1.98 \text{ in}$$

$$\bar{I}_L = 17.3 \text{ in}^4$$

.

For the $b = 8 \text{ in}, h = 1 \text{ in}$ rectangle

$$A_R = bh = 8 \text{ in}^2$$

$$\bar{y}_R = -h/2 = -0.5 \text{ in}$$

$$\bar{I}_R = \frac{bh^3}{12} = \frac{8}{12} = 0.667 \text{ in}^4$$

.

Find the distance to the neutral axis

$$\bar{y} = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{2A_L \bar{y}_L + A_R \bar{y}_R}{2A_L + A_R}$$

$$= \frac{2(4.75)(1.98) + (8)(-0.5)}{2(4.75) + 8}$$

$$\bar{y} = 0.846 \text{ in}$$

3. *Find the Moment of Inertia.*

The distance between the neutral axis and the centroids of the subparts are

$$d_R = |\bar{y} - \bar{y}_R| = |0.846 - (-0.5)| = 1.346 \text{ in}$$

$$d_L = |\bar{y} - \bar{y}_L| = |0.846 - 1.98| = 1.134 \text{ in}$$

The moment of inertia of the rectangle about the x' axis

$$(I'_x)_R = [\bar{I} + Ad^2]_R$$

$$= 0.667 + (8)(1.346)^2$$

$$= 15.16 \text{ in}^4$$

The moment of inertia of one angle about the x' axis

$$(I'_x)_L = [\bar{I} + Ad^2]_L$$

$$= 17.3 + (4.75)(1.134)^2$$

$$= 23.4 \text{ in}^4$$

The moment of inertia of the built up beam about the neutral axis

$$I_{x'} = \sum (I_{x'})_i$$

$$= 2(I_{x'})_L + (I_{x'})_R$$

$$= 2(23.4 \text{ in}^4) + 15.16 \text{ in}^4$$

$$I_{x'} = 61.98 \text{ in}^4$$

10.5 Polar Moment of Inertia

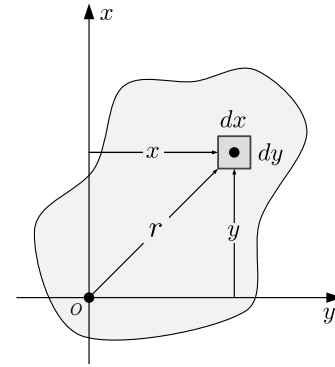
Key Questions

- How are polar moments of inertia similar and different to area moments of inertia about either a horizontal or vertical axis?

The polar moment of inertia is defined by the integral quantity

$$J_O = \int_A r^2 dA, \quad (10.5.1)$$

where r is the distance from the reference point to a differential element of area dA .



The polar moment of inertia describes the distribution of the area of a body with respect to a point in the plane of the body. Alternately, the point can be considered to be where a perpendicular axis crosses the plane of the body. The subscript on the symbol j indicates the point or axis.

There is a particularly simple relationship between the polar moment of inertia and the rectangular moments of inertia. Referring to the figure, apply the Pythagorean theorem $r^2 = x^2 + y^2$ to the definition of polar moment of inertia to get

$$\begin{aligned} J_O &= \int_A r^2 dA \\ &= \int_A (x^2 + y^2) dA \\ &= \int_A x^2 dA + \int_A y^2 dA \\ J_O &= I_x + I_y \end{aligned} \quad (10.5.2)$$

Thinking Deeper 10.5.1 Torsional Stress.

The polar moment of inertia is an important factor in the design of drive shafts. When a shaft is subjected to torsion, it experiences internal distributed shearing forces throughout its cross-section which counteract the external torsional load.

This distributed shearing force is called **shear stress**, and is usually given the symbol tau, τ . Shear stress is zero at the shaft centerline, called the **neutral axis**, and increases linearly with r to a maximum value, $\tau_{\max}$ at

the outside surface where $r = c$, so

$$\tau = \tau_{\max} \frac{r}{c}$$

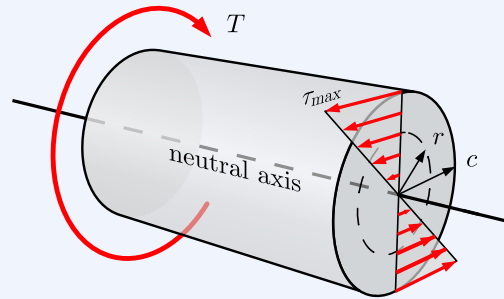


Figure 10.5.2 Section cut through a shaft, showing the shearing stresses developed to withstand an external torsion, T .

The force at any point is $dF = \tau dA$, and the moment dM exerted at any point is dF times the moment arm, r . The total moment is the integral of this quantity over the area of the cross section, and is proportional to the polar moment of inertia.

$$\begin{aligned} T &= \int dM \\ &= \int r dF \\ &= \int_A r \tau_{\max} \frac{r}{c} dA \\ &= \frac{\tau_{\max}}{c} \int_A r^2 dA \\ T &= \frac{\tau_{\max}}{c} J_O \\ \tau_{\max} &= \frac{Tc}{J_O} \end{aligned}$$

This is the relationship between the maximum stress in a circular shaft, the applied torque T , and the geometric properties of the shaft J_O and c .

10.6 Radius of Gyration

The radius of gyration is an alternate way of expressing the distribution of area away from an axis which combines the effects of the moments of inertia and cross sectional area.

The radius of gyration can be thought of as the radial distance to a thin strip which has the same area and the same moment of inertia around a specific

axis as the original shape. Compared to the moment of inertia, the radius of gyration is easier to visualize since it's a distance, rather than a distance to the fourth power.

The radius of gyration, k and the corresponding moment of inertia I are related, and both must refer to the same axis. If one is known, the other is easily found.

The radius of gyration with respect to the x and y axes and the origin are given by these formulas

$$k_x = \sqrt{\frac{I_x}{A}} \quad k_y = \sqrt{\frac{I_y}{A}} \quad k_o = \sqrt{\frac{J_o}{A}} \quad (10.6.1)$$

In engineering design, the radius of gyration is used to determine the stiffness of structural columns and estimate the critical load which will initiate column buckling.

Example 10.6.1

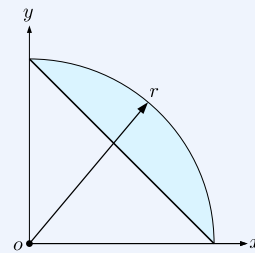
How are k_x , k_y , and k_o related to each other?

Solution. Start with (10.5.2)

$$\begin{aligned} J_o &= I_x + I_y && \text{divide each term by } A \\ \frac{J_o}{A} &= \frac{I_x}{A} + \frac{I_y}{A} && \text{apply definitions of } k^2 \\ k_o^2 &= k_x^2 + k_y^2 \end{aligned}$$

Example 10.6.2

Determine the polar moment of inertia and the radius of gyration of the shaded area. Give answer in terms of r , to three significant digits.



Solution. The shape consists of a quarter circle of radius r with a triangle removed. For the triangle, $b = h = r$.

$$\begin{aligned} A &= \frac{\pi r^2}{4} - \frac{(r)(r)}{2} = 0.285 r^2 \\ I_x &= \frac{\pi r^4}{16} - \frac{(r)(r^3)}{12} = 0.113 r^4 \\ I_y &= I_x \text{ by symmetry.} \\ J_o &= I_x + I_y = 2I_x = 0.226 r^4 \end{aligned}$$

$$k_o = \sqrt{\frac{J_o}{A}} = \sqrt{\frac{0.226 r^4}{0.285 r^2}} = 0.890 r$$

10.7 Products of Inertia

Key Questions

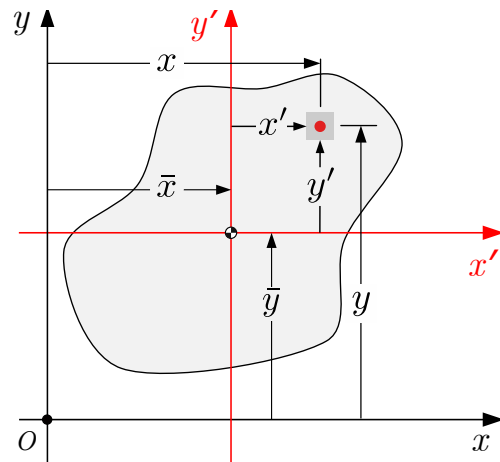
- Why do we need to quantify the product of inertia for beams?
- Why is the product of inertia of a symmetrical cross section zero?

The **product of inertia** is another integral property of area, and is defined as

$$I_{xy} = \int_A xy \, dA. \quad (10.7.1)$$

The parallel axis theorem for products of inertia is

$$I_{xy} = \bar{I}_{x'y'} + A\bar{x}\bar{y}. \quad (10.7.2)$$



Unlike the rectangular moments of inertia, which are always positive, the product of inertia may be either positive, negative, or zero, depending on the object's shape and the orientation of the coordinate axes. The product of inertia will be zero for symmetrical objects when a coordinate axis is also an axis of symmetry.

If the product of inertia is not zero it is always possible to rotate the coordinate system until it is, in which case the new coordinate axes are called the **principle axes**. When the coordinate axes are oriented in the principle directions, the centroidal moments of inertia are maximum about one axis and minimum about the other, but neither is necessarily zero. The principle directions determine the best way to orient a beam to for maximum stiffness, and how much asymmetrical beams, like channels and angles, will twist when a load is applied.

10.8 Mass Moment of Inertia

You may recall from physics the relationship

$$T = I\alpha.$$

This formula is the rotational analog of Newton's second law $F = ma$. Here, the I represents the **mass moment of inertia**, which is the three-dimensional measure of a rigid body's resistance to angular acceleration around an axis. Mass moment of inertia plays the same role for rotational acceleration as *mass* does for linear acceleration.

Mass moment of inertia is defined by an integral equation identical to (10.1.3), except that the differential area dA is replaced with a differential element of mass, dm . The integration is conducted over a three dimensional physical object instead of a two dimensional massless area.

The units of mass moment of inertia are $[\text{mass}][\text{length}]^2$, in contrast to area moment of inertia's units of $[\text{length}]^4$.

Mass moments of inertia are covered in more detail and used extensively in the study of rigid body kinetics in *Engineering Dynamics*.

10.9 Exercises (Ch. 10)

Chapter 10 Exercises

Common Shapes	0/130
Basic Shapes from the MOI Table	0/50
	Not attempted
Rectangles	0/20
	Not attempted
Parallel Axis Theorem: common shapes	0/10
	Not attempted
Rectangles and triangles	0/30
	Not attempted
Semi and quarter circles	0/20
	Not attempted
Composite Shapes	0/100
Triangle and rectangles	0/30
	Not attempted
Composite shape: U or T	0/20
	Not attempted
Semicircle and hole	0/20
	Not attempted
Centroidal Moment of Inertia: T shape	0/20
	Not attempted
Box beam	0/10
	Not attempted
Standard Sections	0/70
Built-up beam with angles	0/30
	Not attempted
2 angles forming box beam	0/20
	Not attempted
2 channels and 2 plates	0/20
	Not attempted

Display options
Done

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In Context